

FROM CHARACTERISTIC CLASSES TO CHERN-WEIL THEORY

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1. INTRODUCTION

In this paper, we will talk about characteristic classes and Chern-Weil theory. Theory of characteristic classes is a powerful tool in algebraic topology which can be applied to many branches of mathematics. In short, a characteristic class assigns a vector bundle specific cohomology classes of the base space to distinguish different vector bundles.

There are many characteristic classes in different situations including Stiefel-Whitney class, Chern class, Euler class, Pontrjagin class and so on. Particularly, Chern class is named after the great Chinese mathematician, Shiing-Shen Chern. He not only extended Stiefel-Whitney class to complex vector bundles to obtain Chern class, but also used connections on vector bundles to create Chern class geometrically, and these two coincide. The latter one was developed by Andrew Weil further more and is now so-called Chern-Weil theory.

To start with, we will take an overall view of the history of characteristic classes and Chern-Weil theory. Then we discuss Stiefel-Whitney class, Chern class, Euler class, Pontrjagin class in detail and cover some beautiful applications. Afterwards,

we introduce connection and curvature on vector bundles, and then use the Chern-Weil homomorphism to define Chern class geometrically, and finally give the results that these two construction coincide.

2. HISTORY

We talk about some history of fiber bundles first.

2.1. Early development of fiber bundles.

The research on tangent vector fields can be dated back to Poincaré in 1880. He considered the vector field on a submanifold embedded in $\mathbb{R}^N$ by intuition. But if the manifold is intrinsically defined, this method will fail. Whitney was the first to propose that the **tangent vector** can be viewed as a derivation in his famous paper [2] in 1935, though not rigorously. Until 1946, Chevalley completed the definition of tangent vectors as linear functionals.

The first person to use the word "**fiber**" might be Seifert in his paper [3]. He studied 3-dimensional manifolds that can be decomposed into S^1 -fibers. After that, Whitney proposed the concept of **fiber bundle** in [4] in 1935, where he called it "sphere spaces" and studied sphere bundles actually. His definition has been used until now: given trivializations, define the transformation map and require them to satisfy the cocycle condition so that we can attach all fibers together. Moreover, he discussed examples of tangent bundles and normal bundles of submanifolds embedded in Euclidean spaces.

After Whitney's definition of fiber bundles, the automorphism group of fiber F , which is now so-called **structure group**, was valued by mathematicians. The French mathematician Ehresmann defined the important concepts of **principal bundles** and **associated bundles** with Feldbau in their paper [5]. As examples, he found the Hopf fibrations $S^3 \rightarrow S^2$, $S^7 \rightarrow S^4$ and $S^{15} \rightarrow S^8$ are all principal bundles with structure group S^1 , S^3 and S^7 respectively.

Other important concepts in vector bundles include **induced bundles**(which is now called **pullback bundles**) and **sections**, which were all proposed by Whitney during 1930s. Generally, a section of the tangent bundle of a manifold, that is, a global vector field, may not exist, such as the sphere S^2 . Hence *sections are some detection to the triviality of the vector bundle*. In particular, a principal bundle is trivial if and only if it has a global section. If we start from a section on a subspace A of the base space B , then the question turns into when this section can be extended to a global section on B ? This is the origin of **characteristic classes** and **obstruction theory**.

2.2. About characteristic classes.

Theory of characteristic classes originated from research on existence of singularities of vector fields. In 1881, Poincaré defined the index of a isolated singularity of a C^1 -vector field on S^2 and proved the sum of indices are 2. Afterwards, Brouwer only assumes the vector field on S^2 is continuous and proved that it must have a singularity in 1909. Then in 1911, Brouwer wrote a paper [6] about the **degree** of a map. He discussed on the continuous vector field of at most finite singularities on S^n and proved that the sum of indices is 2 when n is even and 0 when n is odd. However, his proof is very obscure and hard to convinced. Finally in 1925, Hopf extended Brouwer's results to any compact manifolds and proved the famous Hopf index theorem:

Theorem 2.2.1 (Hopf index theorem). *The sum of the indices of a vector field on a compact oriented manifold M is the Euler characteristic of M . That is,*

$$\sum_p \text{ind}(p) = \chi(M)$$

E. Stiefel is a student of Hopf. Inspired by his teacher, he extended Hopf's research on vector fields in his doctor paper [7]. Given a smooth n -dimensional manifold M , for $m \leq n$, are there m nowhere linearly dependent vector fields on M ? If $m = n$, and there exist such n vector fields, then we generally call M **parallelizable**.

To solve the problem, Stiefel defined the now so-called **Stiefel manifolds** $V_{n,m}$ and $V_{n,m}^0$. $V_{n,m}$ consists of all sets of m linearly independent vectors in $\mathbb{R}^n$, and $V_{n,m}^0 \subseteq V_{n,m}$ requires the vectors to be orthonormal. Using Gram-Schmidt procedure, $V_{n,m}^0$ is a deformation retract of $V_{n,m}$. If we use the language of homotopy theory, he proved that

Proposition 2.2.2 (Stiefel). *When $i < n - m$, $\pi_i(V_{n,m}) = 0$. When $m = 1$ or $n - m$ is even, $\pi_{n-m}(V_{n,m}) = \mathbb{Z}$. When $m \neq 1$ and $n - m$ is odd, $\pi_{n-m}(V_{n,m}) = \mathbb{Z}_2$.*

At that time the concept of cohomology was just defined, so Stiefel didn't use the language of cohomology. Instead, he considered barycentric subdivision of the dual complex of a triangulation T of a manifold M and considered some "chains". If we use the language of homology and cohomology, what he did is that he defined a cohomology class, and then used Poincaré duality to obtain a homology class $F \in H_{m-1}(M; \pi_{n-m}(V_{n,m}))$. Stiefel used his characteristic class to prove the following theorem:

Theorem 2.2.3 (Stiefel). *Any compact orientable smooth 3-dimensional manifold is always parallelizable.*

Whitney also discussed any sphere bundle $E \rightarrow B$ and obtained the definition of characteristic classes in his paper [2] in 1935. He also derived the now so-called **Stiefel-Whitney classes** $w_r \in H^r(B; \mathbb{Z}_2)$.

The method that Stiefel and Whitney used can be extended to general obstruction theory actually. Suppose ξ is a rank n vector bundle. For any $k \leq n$, denote the associated bundle with fiber $V_{n,k}^0$ by $V_k(\xi)$. Since $V_{n,k}^0$ is $n - k - 1$ connected, we can define sections of $V_k(\xi)$ on $n - k$ skeleton B^{n-k} of B . If we want to extend the section onto B^{n-k+1} , then we get obstruction cocycle, whose cohomology class is called **obstruction class**

$$\mathfrak{o}_{n-k+1}(\xi) \in H^{n-k+1}(B; \{G_x\})$$

where $\{G_x\}$ consists of local coefficients $\{\pi_{n-k}(V_{n-k}^0(x))\}_{x \in B}$. If we use the natural map $\pi_{n-k}(V_{n-k}^0(x)) \rightarrow \mathbb{Z}_2$, then we get the Stiefel-Whitney class

$$w_{n-k+1}(\xi) \in H^{n-k+1}(B; \mathbb{Z}_2)$$

Another important consequence by Whitney is the following product theorem in his paper [8][9] in 1940.

Theorem 2.2.4 (Product theorem). *Suppose ξ_1, ξ_2 are two vector bundles, then*

$$w_k(\xi_1 \oplus \xi_2) = \sum_{i=0}^k w_i(\xi_1) \smile w_{k-i}(\xi_2)$$

However, he wrote in his paper that the proof is difficult and gave only a little description. The first proof of the product theorem was given by Wu Wen-Tsün and Shiing-Shen Chern.

Stiefel-Whitney class has many applications. A vector bundle is orientable if and only if $w_1(\xi) = 0$. It can also solve the problem of immersion theory, for example, when $\mathbb{RP}^n$ can be immersed into $\mathbb{R}^m$. Applied to algebra, it solves the existence of division algebra.

Pontryagin was a great Soviet mathematician. He lost his eyesight completely when he is 14. However, he was the first to discover that the Stiefel-Whitney class can be defined by pulling back the cohomology class of Grassmannian (which is one of the classifying spaces). So he studied many properties of Grassmannian in detail and defined the **Pontryagin classes** $p_k(\xi) \in H^{4k}(B, \mathbb{Z})$ and **Euler classes** $e(\xi) \in H^m(B; \mathbb{Z})$ using the same method.

Shiing-Shen Chern extended Pontryagin's results to complex vector bundles in 1945. He took complex Stiefel manifolds and complex Grassmannian, defined **Chern classes** $c_k(\xi) \in H^{2k}(B; \mathbb{Z})$ for any complex vector bundles ξ by pullback. Wu Wen-Tsün also worked with him this time, and proved the product theorem for Chern class:

Theorem 2.2.5 (Product theorem). *Suppose ξ_1, ξ_2 are two vector complex bundles, then*

$$c_k(\xi_1 \oplus \xi_2) = \sum_{i=0}^k c_i(\xi_1) \smile c_{k-i}(\xi_2)$$

Moreover, he gave the connection between Pontryagin class and Chern class:

Proposition 2.2.6. *If ξ is a real vector bundle, then*

$$p_j(\xi) = (-1)^j c_{2j}(\xi \otimes \mathbb{C})$$

There are some other advanced consequences of characteristic classes, and we will discuss selectively in the following sections. Here is a sketch:

A. Suppose ξ is a real vector bundle of rank n with its Stiefel-Whitney classes $w_r(\xi)$ for $1 \leq r \leq n$. Then we define $w_0(\xi) = 1 \in H^0(B; \mathbb{Z}_2)$ and $w_r(\xi) = 0 \in H^r(B; \mathbb{Z}_2)$ for $r > n$. Denote $w(\xi) = 1 + w_1(\xi) + \cdots + w_n(\xi)$, we call it the **total Stiefel-Whitney class**. Then the product theorem can be written in the form:

$$w(\xi_1 \oplus \xi_2) = w(\xi_1) \smile w(\xi_2)$$

Suppose ξ is a real vector bundle of rank n with its Pontryagin classes $p_r(\xi)$ for $1 \leq r \leq n/2$. Then we define $p_0(\xi) = 1$ and $p_r(\xi) = 0$ for $r > n/2$. Denote $p(\xi) = 1 + p_1(\xi) + \cdots + p_{[n/2]}(\xi)$, we call it the **total Pontryagin class**. Then the product theorem can be written in the form:

$$2(p(\xi_1 \oplus \xi_2) - p(\xi_1) \smile p(\xi_2)) = 0$$

Suppose ξ is a complex vector bundle of rank n with its Chern classes $c_r(\xi)$ for $1 \leq r \leq n$. Then we define $c_0(\xi) = 1$ and $c_r(\xi) = 0$ for $r > n$. Denote $c(\xi) = 1 + c_1(\xi) + \cdots + c_n(\xi)$, we call it the **total Chern class**. Then the product theorem can be written in the form:

$$c(\xi_1 \oplus \xi_2) = c(\xi_1) \smile c(\xi_2)$$

B. In 1950, Wu Wen-Tsün used Steenrod squares $Sq = Sq^0 + Sq^1 + \cdots$ to prove the following consequence in [10]:

Theorem 2.2.7 (Wu). *For any compact manifold M , there exists a unique cohomology class $v \in H^*(M; \mathbb{Z}_2)$, $v = v_1 + v_2 + \cdots + v_n$, $v_k \in H^k(M; \mathbb{Z}_2)$, such that for all $x \in H^*(M; \mathbb{Z}_2)$*

$$v \smile x = Sq(x)$$

v is called the **Wu class**.

The Wu class and Stiefel Whitney class have the following relation:

Theorem 2.2.8 (Wu's formula).

$$w(M) = Sq(v)$$

or,

$$w_k(M) = \sum_{i+j=k} Sq^i(v_j)$$

Moreover, in [11], Wu proved the Stiefel-Whitney classes have the following relation:

Theorem 2.2.9 (Wu's relation).

$$Sq^k(w_m) = w_k w_m + \binom{k-m}{1} w_{k-1} w_{m+1} + \cdots + \binom{k-m}{k} w_0 w_{m+k}$$

C. In 1956, Hirzebruch proposed the axiom systems for characteristic classes. The axiom system for Stiefel-Whitney classes is:

Axiom 1 For any real vector bundle ξ with base space B and rank n and for all $i \leq 0$, there exists a Stiefel-Whitney class $w_i(\xi) \in H^i(B; \mathbb{Z}_2)$, where $w_0(\xi) = 1$ and $w_i(\xi) = 0$ when $i > n$. Denote $\sum_i w_i(\xi)$ by $w(\xi)$.

Axiom 2 (Naturality) $w(f^*\xi) = f^*(w(\xi))$.

Axiom 3 (Whitney product theorem) $w(\xi \oplus \eta) = w(\xi) \smile w(\eta)$.

Axiom 4 (Normalization) Suppose η_n is the canonical line bundle of $\mathbb{RP}^n$, then $w(\eta_n) = 1 + h_n$, where h_n is the generator of $H^1(\mathbb{RP}^n; \mathbb{Z}_2)$.

The axiom system for Chern classes is:

Axiom 1 For any complex vector bundle ξ with base space B and rank n and for all $i \leq 0$, there exists a Chern class $c_i(\xi) \in H^{2i}(B; \mathbb{Z})$, where $c_0(\xi) = 1$ and $c_i(\xi) = 0$ when $i > n$. Denote $\sum_i c_i(\xi)$ by $c(\xi)$.

Axiom 2 (Naturality) $c(f^*\xi) = f^*(c(\xi))$.

Axiom 3 (Whitney product theorem) $c(\xi \oplus \eta) = c(\xi) \smile c(\eta)$.

Axiom 4 (Normalization) Suppose η_n is the canonical line bundle of $\mathbb{CP}^n$, then $c(\eta_n) = 1 + h_n$, where c_n is the generator of $H^2(\mathbb{CP}^n; \mathbb{Z})$.

D. In 1944, Pontryagin proved the following consequence in his paper [12]:

Theorem 2.2.10 (Pontryagin). *For any compact orientable manifold M , there is a formula*

$$\langle e(TM), [M] \rangle = \chi(M)$$

This is also why $e(TM)$ is called the Euler class. If we use the de Rham cohomology, then the formula can be written in integral and it can also be used to prove the Hopf index theorem 2.2.1, see [13].

E. The Gysin sequence can be stated using Euler class:

Theorem 2.2.11 (Gysin sequence). *Suppose ξ is a oriented rank n real vector bundle with base space B . E_0 consists of all nonzero vectors in E . $\pi_0 = \pi|_{E_0}$. Suppose $e \in H^n(B; \mathbb{Z})$ is the Euler class of ξ , then there exists a long exact sequence*

$$\cdots \longrightarrow H^i(B; \mathbb{Z}) \xrightarrow{\smile e} H^{i+n}(B; \mathbb{Z}) \xrightarrow{\pi_0^*} H^{i+n}(E_0; \mathbb{Z}) \longrightarrow H^{i+1}(B; \mathbb{Z}) \longrightarrow \cdots$$

2.3. Brief history of Chern-Weil theory.

The origin of Chern-Weil theory was dependent on connections on principal bundles. As we discussed before, the concept of principal bundles was defined by Ehresmann. In the first half of the 20th century, there were various definitions of **connection** in geometry. He was a student of Cartan, and he learned Cartan's connections from a global point of view as well as being influenced by Lie and Vessiot. In 1951, he defined an **infinitesimal connection**, which is now so-called **Ehresmann connection**, on a locally trivial fiber bundle to be a distribution of n -contact elements which is transverse to the fibers and satisfies the path lifting property. In the case of a principal G -bundle, Ehresmann adds the condition that the distribution must be invariant under the right action of G .

In light of Ehresmann's definition, a connection can be seen as a purely geometric object which is equivalent to parallel transport.

Chern's work on Chern class originated from his proof of Gauss-Bonnet-Chern formula. Chern considered it as his best work ever. He extended Gauss-Bonnet theorem to higher dimensions by considering curvature forms, using one vector field and looking at its set of zero to find the Euler characteristic of the manifold, see [17]. He got inspiration from the proof, and due to previous mathematicians' work on characteristic classes, he finally came up with the idea of Chern class.

For a complex vector bundle, Chern defined Chern classes in three different ways: by obstruction theory, by Schubert cells, and by curvature forms of a connection on a complex vector bundle. And he proved their equivalence. In particular, the third way in his paper [18] also laid the foundation for performing Hermitian geometry on complex manifolds. If Ω is the curvature form of a complex vector bundle, he defined Chern classes to be

$$\det \left(I - \frac{i}{2\pi} \Omega \right) = 1 + c_1(\Omega) + \cdots + c_q(\Omega)$$

Moreover, he defined the Chern-Simons form which played a crucial role in three dimension manifolds, in string theory and so on. After his fundamental paper on Chern classes in 1946, Chern explored more details on the multiplicative structure of the characteristic classes.

However, Chern wasn't aware that the cohomology classes were independent of choices of connections. It was Andrew Weil to complete the proof of independence. Besides, Weil used more algebraic approaches to extended Chern's work to any principal G -bundles and specified the Chern-Weil homomorphism connecting with de Rham cohomology, starting from his work in [19]. With advanced correction and development, Chern-Weil theory walked to mature gradually. It has deep influence on physics, especially on Yang-Mills theory and topological quantum field theory.

3. CHARACTERISTIC CLASSES

This section will give a detailed discussion on characteristic classes and their applications. The main reference is [20].

3.1. Stiefel-Whitney class.

In the following we use ξ to denote a vector bundle $\pi : E \rightarrow B$ without specification. We first use the axiom system to define Stiefel-Whitney classes, and then prove its existence and uniqueness.

Definition 3.1.1 (Stiefel-Whitney classes). *We give the following axiom system:*

Axiom 1 *To each vector bundle ξ there corresponds a sequence of cohomology classes*

$$w_i(\xi) \in H^i(B; \mathbb{Z}_2), \quad i = 0, 1, 2, \dots$$

*called the **Stiefel-Whitney class of ξ** . The class $w_0(\xi)$ is equal to the unit element*

$$1 \in H^0(B; \mathbb{Z}_2)$$

and $w_i(\xi)$ equals zero for $i > n$ if ξ is an n -plane bundle.

Axiom 2 (Naturality) *If $f : B(\xi) \rightarrow B(\eta)$ is covered by a bundle map from ξ to η , then*

$$w_i(\xi) = f^*(w_i(\eta))$$

Axiom 3 (Whitney product theorem) *If ξ and η are vector bundles over the same base space, then*

$$w_k(\xi \oplus \eta) = \sum_{i=0}^k w_i(\xi) \smile w_{k-i}(\eta)$$

Axiom 4 *For the line bundle γ_1^1 over the circle $\mathbb{RP}^1$, the Stiefel-Whitney class $w_1(\gamma_1^1)$ is non-zero.*

Definition 3.1.2. $H^\Pi(B; \mathbb{Z}_2)$ *denote the ring consisting of all formal infinite series*

$$a = a_0 + a_1 + a_2 + \dots$$

with $a_i \in H^i(B; \mathbb{Z}_2)$.

Definition 3.1.3. *The **total Stiefel-Whitney class** of an n -plane bundle ξ over B is defined to be the element*

$$w(\xi) = 1 + w_1(\xi) + w_2(\xi) + \dots + w_n(\xi) + 0 + \dots$$

of the ring $H^\Pi(B; \mathbb{Z}_2)$.

Interlude 1: Existence of Stiefel-Whitney classes

Generally, the above axioms are enough for us to explore properties of Stiefel-Whitney classes and give some applications. And the proof of existence is somehow technique and based on many nontrivial consequences, hence we make it an interlude.

First we need two tools: Thom isomorphism and Steenrod squares.

For any n -plane bundle ξ with total space E , base space B and projection π , we denote by E_0 the set of all non-zero elements of E , and by F_0 the set of all non-zero elements of a typical fiber $F = \pi^{-1}(b)$.

In the following, the coefficient group $\mathbb{Z}_2$ is used.

Theorem 3.1.4 (Thom isomorphism, see [13]). *The group $H^i(E, E_0)$ is zero for $i < n$, and $H^n(E, E_0)$ contains a unique class u such that for each fiber $F = \pi^{-1}(b)$ the restriction*

$$u|_{(F, F_0)} \in H^n(F, F_0)$$

*is the unique non-zero class in $H^n(F, F_0)$. Furthermore the correspondence $x \mapsto x \smile u$ defines an isomorphism $H^k(E) \rightarrow H^{k+n}(E, E_0)$ for every k . (We call u the **fundamental cohomology class**.)*

Remark 3.1.5. *If the vector bundle is orientable, then the Thom isomorphism also holds for $\mathbb{Z}$ coefficient.*

Definition 3.1.6. *The **Thom isomorphism** $\phi : H^k(B) \rightarrow H^{k+n}(E, E_0)$ is defined to be the composition of the two isomorphisms*

$$H^k(B) \xrightarrow{\pi^*} H^k(E) \xrightarrow{\smile u} H^{k+n}(E, E_0)$$

Theorem 3.1.7 (Steenrod squares, see [21]). *There is an operation in $H^*(E, E_0)$ called **Steenrod squares** defined as follows: for each pair $Y \subseteq X$ of spaces and each pair n, i of non-negative integers, there is defined an additive homomorphism*

$$Sq^i : H^n(X, Y) \rightarrow H^{n+i}(X, Y)$$

satisfying the following properties:

- (1) (**Naturality**) *If $f : (X, Y) \rightarrow (X', Y')$ then $Sq^i \circ f^* = f^* \circ Sq^i$.*
- (2) *If $a \in H^n(X, Y)$, then $Sq^0(a) = a$, $Sq^n(a) = a \smile a$ and $Sq^i(a) = 0$ for $i > n$.*
- (3) (**The Cartan formula**) *The identity*

$$Sq^k(a \smile b) = \sum_{i+j=k} Sq^i(a) \smile Sq^j(b)$$

is valid whenever $a \smile b$ is defined.

Using the above consequences, we can define the **Stiefel-Whitney class** $w_i(\xi) \in H^i(B)$ to be

$$w_i(\xi) = \phi^{-1} Sq^i \phi(1).$$

In other words, $w_i(\xi)$ is the unique cohomology class in $H^i(B)$ such that $\phi(w_i(\xi)) = \pi^* w_i(\xi) \smile u$ is equal to $Sq^i \phi(1) = Sq^i(u)$.

It is convenient to introduce the **total squaring operation**

$$Sq(a) = 1 + Sq^1(a) + Sq^2(a) + \cdots + Sq^n(a)$$

for $a \in H^n(X, Y)$. Then the Cartan formula can be expressed as

$$Sq(a \smile b) = Sq(a) \smile Sq(b)$$

In terms of this total squaring operation, the total Stiefel-Whitney class of a vector bundle is determined by

$$w(\xi) = \phi^{-1} Sq(\phi(1)) = \phi^{-1} Sq(u)$$

Now let's verify the four axioms for Stiefel-Whitney classes:

Axiom 1. Using the property (2) of Sq , we immediately get the result.

Axiom 2. Any bundle map $f : \xi \rightarrow \xi'$ induces a map $g : (E, E_0) \rightarrow (E', E'_0)$.

Moreover, if u' is the fundamental cohomology class in $H^n(E', E'_0)$, then $g^*(u')$ is equal to the class $u \in H^n(E, E_0)$. Hence by the property (1) of Sq

$$\begin{aligned} f^*(w_i(\eta)) &= f^*(\phi'^{-1}Sq^i(u')) = \phi^{-1}(g^*Sq^i(u')) \\ &= \phi^{-1}(Sq^i g^*(u')) = \phi^{-1}(Sq^i(u)) = w_i(\xi) \end{aligned}$$

as required.

Axiom 3. Let us first compute the Stiefel-Whitney classes of a product $\xi'' = \xi \times \xi'$, with projection map $\pi \times \pi' : E \times E' \rightarrow B \times B'$. Consider the fundamental classes

$$u \in H^m(E, E_0), \quad u' \in H^n(E', E'_0)$$

of ξ and ξ' . Then

$$u \times u' \in H^{m+n}(E \times E', E \times E'_0 \cup E_0 \times E')$$

is defined. Note that $E \times E'_0 \cup E_0 \times E'$ is exactly E''_0 . In fact we **claim** that $u \times u'$ is exactly the fundamental class $u'' \in H^{m+n}(E'', E''_0)$. To prove this, it suffices to show that the restriction

$$u \times u'|_{(F'', F''_0)}$$

is the non-zero cohomology class in $H^{m+n}(F'', F''_0)$ for every fiber $F'' = F \times F'$ of ξ'' . But this restriction is evidently equal to the cross product of $u|_{(F, F_0)}$ and $u'|_{(F', F'_0)}$, and hence is non-zero. Then the Thom isomorphisms for ξ , ξ' and ξ'' are related by

$$\begin{aligned} \phi(a) \times \phi'(b) &= (\pi^*(a) \smile u) \times (\pi'^*(b) \smile u') = (\pi^*(a) \times \pi'^*(b)) \smile (u \times u') \\ &= \pi^*(a \times b) \smile u'' = \phi''(a \times b) \end{aligned}$$

where there is no sign since we are working with $\mathbb{Z}_2$ coefficients.

The total Stiefel-Whitney class of ξ'' can now be computed as follows:

$$\begin{aligned} \phi''(w(\xi'')) &= Sq(u'') = Sq(u \times u') = Sq(u) \times Sq(u') \\ &= \phi(w(\xi)) \times \phi'(w(\xi')) = \phi''(w(\xi) \times w(\xi')) \end{aligned}$$

and then applying $(\phi'')^{-1}$ to both sides, we get

$$w(\xi'') = w(\xi) \times w(\xi')$$

Finally, note that the Whitney sum of vector bundles can be obtained by using the diagonal map $B \rightarrow B \times B$ to pull back the product bundle. Hence by Axiom 2, we get

$$w(\xi \oplus \xi') = w(\xi) \smile w(\xi')$$

Axiom 4. Let γ_1^1 be the canonical twisted line bundle over the circle $\mathbb{RP}^1$. Then the space of vectors of length ≤ 1 in the total space $E = E(\gamma_1^1)$ is evidently a Möbius band M , bounded by a circle S . Since M is a deformation retract of E and S a deformation retract of E_0 , we have

$$H^*(M, S) \cong H^*(E, E_0)$$

On the other hand if we embed a 2-cell D^2 in $\mathbb{RP}^2$, then the closure of $\mathbb{RP}^2 \setminus D^2$ is homeomorphic to M . Using the Excision theorem, we get

$$H^*(M, S) \cong H^*(\mathbb{RP}^2, D^2)$$

Hence there are natural isomorphisms

$$H^i(E, E_0) \longrightarrow H^i(M, S) \longleftarrow H^i(\mathbb{RP}^2, D^2) \longrightarrow H^i(\mathbb{RP}^2)$$

for every dimension $\neq 0$. The fundamental cohomology class $u \in H^1(E, E_0)$ is certainly not zero. Hence it must correspond to the generator $a \in H^1(RP^2)$ under the composite isomorphism. Hence $Sq^1(u) = u \smile u$ must correspond to $Sq^1(a) = a \smile a$. But $a \smile a \neq 0$, so it follows that

$$w_1(\gamma_1^1) = \phi^{-1}Sq^1(u) \neq 0$$

This concludes the verification of the four axioms.

Now let's come back to the discussion on properties of Stiefel-Whitney classes. Axiom 2 has immediate consequences as follows:

Proposition 3.1.8. *If ξ is isomorphic to η , then $w_i(\xi) = w_i(\eta)$.*

Proposition 3.1.9. *If ε is a trivial bundle, then $w_i(\varepsilon) = 0$ for $i > 0$.*

Proof. If ε is trivial, then there exists a bundle map from ε to a vector bundle over a point. $\square$

Combining this result with the Whitney product theorem, one obtains:

Proposition 3.1.10. *If ε is trivial, then $w_i(\varepsilon \oplus \eta) = w_i(\eta)$.*

Proposition 3.1.11. *If ξ is an $\mathbb{R}^n$ -bundle with a Euclidean metric which possesses a nowhere zero cross-section, then $w_n(\xi) = 0$. If ξ possesses k cross-sections which are nowhere linearly dependent, then*

$$w_{n-k+1}(\xi) = w_{n-k+2}(\xi) = \cdots = w_n(\xi) = 0$$

Proof. ξ splits as a Whitney sum $\varepsilon \oplus \varepsilon^\perp$ where ε is trivial and $\varepsilon^\perp$ has dimension $n - k$. Then it follows from last proposition. $\square$

Now let's consider the special case that the Whitney sum $\xi \oplus \eta$ is trivial, then we have

$$w(\xi)w(\eta) = 1 \in H^1(B; \mathbb{Z}_2)$$

If we denote the inverse of $w \in H^1(B; \mathbb{Z}_2)$ by $\bar{w}$, then

$$w(\eta) = \bar{w}(\xi)$$

If we write it out,

$$\begin{aligned} \bar{w} &= (1 + (w_1 + w_2 + w_3 + \cdots))^{-1} \\ &= 1 - (w_1 + w_2 + w_3 + \cdots) + (w_1 + w_2 + \cdots)^2 - (w_1 + w_2 + \cdots)^3 + \cdots \\ &= 1 - w_1 + (w_1^2 - w_2) + (-w_1^3 + 2w_1w_2 - w_3) + \cdots \end{aligned}$$

Since the signs are irrelevant under $\mathbb{Z}_2$ coefficients, this leads to the precise expression $(i_1 + \cdots + i_k)!/i_1! \cdots i_k!$ for the coefficient of $w_1^{i_1}w_2^{i_2} \cdots w_k^{i_k}$ in $\bar{w}$.

There is a special case that has many applications:

Lemma 3.1.12 (Whitney duality theorem). *If τ_M is the tangent bundle of a manifold in Euclidean space and ν is the normal bundle, then*

$$w_i(\nu) = \bar{w}_i(\tau_M)$$

Example 3.1.13. *For the tangent bundle τ of the unit sphere S^n , the class $w(\tau) = w(S^n) = 1$. This is because the normal bundle ν is trivial and $w(\tau)w(\nu) = 1$.*

In the following we consider bundles over the projective space $\mathbb{RP}^n$. Recall that the cohomology ring of $\mathbb{RP}^n$ with coefficient $\mathbb{Z}_2$ is given by

$$H^*(\mathbb{RP}^n; \mathbb{Z}_2) = \mathbb{Z}[a]/(a^{n+1})$$

Example 3.1.14. *The total Stiefel-Whitney class of the canonical line bundle γ_n^1 over $\mathbb{RP}^n$ is given by*

$$w(\gamma_n^1) = 1 + a$$

This is because the standard inclusion $j : \mathbb{RP}^1 \rightarrow \mathbb{RP}^n$ is clearly covered by a bundle map from γ_1^1 to γ_n^1 . Therefore

$$j^*w(\gamma_n^1) = w(\gamma_1^1)$$

Example 3.1.15. *The line bundle γ_n^1 over $\mathbb{RP}^n$ is contained as a subbundle in the trivial bundle ε_{n+1} . Let $\gamma_n^\perp$ denote the orthogonal complement of γ_n^1 in ε_{n+1} . (The total space $E(\gamma_n^\perp)$ consists of all pairs*

$$(\{\pm x, v\}) \in \mathbb{RP}^n \times \mathbb{R}^{n+1}$$

with v perpendicular to x .) Then

$$w(\gamma_n^\perp) = \overline{w}(\gamma_n^1) = (1 + a)^{-1} = 1 + a + a^2 + \cdots + a^n$$

Now we consider the tangent bundle of $\mathbb{RP}^n$.

Lemma 3.1.16. *The tangent bundle τ of $\mathbb{RP}^n$ is isomorphic to $\text{Hom}(\gamma_n^1, \gamma_n^\perp)$.*

Proof. Let L be a line through the origin in $\mathbb{R}^{n+1}$, intersecting S^n in the points $\pm x$, and let $L^\perp \subseteq \mathbb{R}^{n+1}$ be the complementary n -plane. Let $f : S^n \rightarrow \mathbb{RP}^n$ denote the canonical map, $f(x) = \{\pm x\}$. Note that the two tangent vectors (x, v) and $(-x, -v)$ in TS^n both have the same image under the map

$$df : TS^n \rightarrow T\mathbb{RP}^n$$

Thus the tangent manifold $T\mathbb{RP}^n$ can be identified with the set of all pairs $\{(x, v), (-x, -v)\}$ satisfying

$$x \cdot x = 1, \quad x \cdot v = 0$$

But each such pair determines, and is determined by, a linear map

$$\ell : L \rightarrow L^\perp$$

where $\ell(x) = v$. Thus the tangent space of $\mathbb{RP}^n$ at $\{\pm x\}$ is canonically isomorphic to the vector space $\text{Hom}(L, L^\perp)$. It follows that tangent vector bundle τ is canonically isomorphic to the bundle $\text{Hom}(\gamma_n^1, \gamma_n^\perp)$. $\square$

We cannot derive $w(\mathbb{RP}^n)$ from this lemma because we don't know the Stiefel-Whitney classes of Hom bundle. However, the computation can be carried as follows.

Theorem 3.1.17. *The Whitney sum $\tau \oplus \varepsilon^1$ is isomorphic to the $(n+1)$ -fold Whitney sum $\gamma_n^1 \oplus \gamma_n^1 \oplus \cdots \oplus \gamma_n^1$. Hence the total Stiefel-Whitney class of $\mathbb{RP}^n$ is given by*

$$w(\mathbb{RP}^n) = (1 + a)^{n+1} = 1 + \binom{n+1}{1}a + \binom{n+1}{2}a^2 + \cdots + \binom{n+1}{n}a^n$$

Proof. The bundle $\text{Hom}(\gamma_n^1, \gamma_n^1)$ is trivial since it is a line bundle with a canonical nowhere zero cross-section. Therefore

$$\begin{aligned} \tau \oplus \varepsilon^1 &\cong \text{Hom}(\gamma_n^1, \gamma^\perp) \oplus \text{Hom}(\gamma_n^1, \gamma_n^1) \\ &\cong \text{Hom}(\gamma_n^1, \gamma^\perp \oplus \gamma_n^1) \\ &\cong \text{Hom}(\gamma_n^1, \varepsilon^{n+1}) \\ &\cong \text{Hom}(\gamma_n^1, \varepsilon^1) \oplus \cdots \oplus \text{Hom}(\gamma_n^1, \varepsilon^1) \\ &\cong \gamma_n^1 \oplus \cdots \oplus \gamma_n^1 \end{aligned}$$

Then the Stiefel-Whitney classes of $\mathbb{RP}^n$ follows from the Whitney product theorem. $\square$

Recall the definition of parallelizable:

Definition 3.1.18. A smooth manifold is said to be **parallelizable** if its tangent bundle is trivial.

From the previous theorem, we have

Corollary 3.1.19 (Stiefel). *The class $w(\mathbb{RP}^n)$ is equal to 1 if and only if $n + 1$ is a power of 2. Thus the only projective spaces which can be parallelizable are $\mathbb{RP}^1, \mathbb{RP}^3, \mathbb{RP}^7, \mathbb{RP}^{15}, \dots$.*

Remark 3.1.20. *In fact, the parallelizable projective spaces are only $\mathbb{RP}^1, \mathbb{RP}^3, \mathbb{RP}^7$. One can use spectral sequences to prove this.*

Proof. The identity $(a + b)^2 \equiv a^2 + b^2 \pmod{2}$ implies that

$$(1 + a)^{2^r} \equiv 1 + a^{2^r} \pmod{2}$$

Therefore if $n + 1 = 2^r$, then

$$w(\mathbb{RP}^n) = (1 + a)^{n+1} = 1 + a^{n+1} = 1$$

Conversely, if $n + 1 = 2^r m$ with m odd, $m > 1$, then

$$w(\mathbb{RP}^n) = (1 + a)^{n+1} = (1 + a^{2^r})^m = 1 + ma^{2^r} + \cdots \neq 1$$

$\square$

Theorem 3.1.17 has another very important application to solve which projective spaces can be immersed in the Euclidean space of a given dimension.

If a manifold M of dimension n can be immersed in the Euclidean space $\mathbb{R}^{n+k}$, then the Whitney duality theorem

$$w_i(\nu) = \overline{w}_i(M)$$

implies that the dual Stiefel-Whitney classes $\overline{w}_i(M)$ are zero for $i > k$.

For example, since

$$w(\mathbb{RP}^9) = (1 + a)^{10} = 1 + a^2 + a^8$$

we have

$$\overline{w}(\mathbb{RP}^9) = 1 + a^2 + a^4 + a^6$$

Thus if $\mathbb{RP}^9$ can be immersed in $\mathbb{R}^{9+k}$, then k must be at least 6.

The most interesting results for $\mathbb{RP}^n$ are obtained when n is a power of 2. If $n = 2^r$, then

$$w(\mathbb{RP}^n) = (1 + a)^{n+1} = 1 + a + a^n$$

hence

$$\overline{w}(\mathbb{RP}^n) = 1 + a + a^2 + \cdots + a^{n-1}$$

Thus we proved that:

Theorem 3.1.21. *If $\mathbb{RP}^{2^r}$ can be immersed in $\mathbb{R}^{2^r+k}$, then k must be at least $2^r - 1$.*

On the other hand, we have the well-known Whitney theorem that every smooth compact manifold of dimension n can be immersed in $\mathbb{R}^{2n-1}$. Thus the theorem gives the best estimate for $\mathbb{RP}^{2^r}$.

However, previous argument may not give the best estimate for other $\mathbb{RP}^n$. For example, we have known that $\mathbb{RP}^8$ can't be immersed in $\mathbb{R}^{14}$. But in fact, there is a proof given by James in [22] that $\mathbb{RP}^9$ can't be immersed in $\mathbb{R}^{14}$.

Remark 3.1.22. $\mathbb{RP}^2$ can be immersed in $\mathbb{R}^3$ which is called the Boy's surface, but $\mathbb{RP}^2$ can be embedded in $\mathbb{R}^3$. This can also be proved using Stiefel-Whitney classes, which we will discuss later.

Now we consider a closed, smooth n -dimensional manifold M . Using mod 2 coefficients, by Poincaré duality, there is a unique **fundamental class**

$$\mu_M \in H_n(M; \mathbb{Z}_2)$$

Hence for any cohomology class $v \in H^n(M; \mathbb{Z}_2)$, the **Kronecker index**

$$\langle v, \mu_M \rangle \in \mathbb{Z}_2$$

is defined. We will also use $v[M]$ to denote the index.

Let $r_1, \dots, r_n$ be non-negative integers with $r_1 + 2r_2 + \cdots + nr_n = n$, then for any vector bundle ξ , we have

$$w_1(\xi)^{r_1} \cdots w_n(\xi)^{r_n} \in H^n(\mathbb{Z}_2)$$

Hence we can define

Definition 3.1.23. *The corresponding integer mod 2*

$$w_1^{r_1} \cdots w_n^{r_n}[M]$$

*is called the **Stiefel-Whitney number** of M associated with the monomial $w_1^{r_1} \cdots w_n^{r_n}$.*

As an example, we compute the Stiefel-Whitney numbers of $\mathbb{RP}^n$. Let τ denote the tangent bundle of $\mathbb{RP}^n$. If n is even, then $w_n(\tau) = (n+1)a^n$ is non-zero, and it follows that the Stiefel-Whitney number $w_n[\mathbb{RP}^n] \neq 0$. In particular, if n is a power of 2, then $w(\tau) = 1 + a + a^n$, and it follows that all other Stiefel-Whitney numbers of $\mathbb{RP}^n$ are zero.

On the other hand, if n is odd, say $n = 2k - 1$, then $w(\tau) = (1+a)^{2k} = (1+a^2)^k$, so it follows that $w_j(\tau) = 0$ whenever j is odd. Since every monomial of total dimension $2k - 1$ must contain a factor of odd dimension, it follows that all Stiefel-Whitney numbers of $\mathbb{RP}^{2k-1}$ are zero.

In general, we can derive the Stiefel-Whitney numbers for a particular class of manifolds:

Theorem 3.1.24 (Pontrjagin). *If B is a smooth compact $(n+1)$ -dimensional manifold with boundary equal to M , then the Stiefel-Whitney numbers of M are all zero.*

Proof. Denote the fundamental class of the pair by

$$\mu_B \in H_{n+1}(B, M)$$

where the coefficient group $\mathbb{Z}_2$ is used. The natural homomorphism

$$\partial : H_{n+1}(B, M) \rightarrow H_n(M)$$

maps μ_B to μ_M . For any class $v \in H^n(M)$, note the identity

$$\langle v, \partial \mu_B \rangle = \langle \delta v, \mu_B \rangle$$

where δ is the natural homomorphism $H^n(M) \rightarrow H^{n+1}(B, M)$. Now we consider the tangent bundle τ_B restricted to M and the subbundle τ_M , and it follows that

$$\tau_B|_M \cong \tau_M \oplus \varepsilon^1$$

where the trivial line bundle is spanned by the normal vector field. Hence the Stiefel-Whitney classes of τ_B restricted to M , is equal to the Stiefel-Whitney classes of τ_M . Using the exact sequence

$$H^n(B) \xrightarrow{i^*} H^n(M) \xrightarrow{\delta} H^{n+1}(B, M)$$

it follows that

$$\delta(w_1^{r_1} \cdots w_n^{r_n}) = 0$$

and therefore

$$\langle w_1^{r_1} \cdots w_n^{r_n}, \partial \mu_B \rangle = \langle \delta(w_1^{r_1} \cdots w_n^{r_n}), \mu_B \rangle = 0$$

Thus all Stiefel-Whitney numbers of M are zero. $\square$

The converse is also true, but much harder to prove.

Theorem 3.1.25 (Thom). *If all Stiefel-Whitney numbers of M are zero, then M can be realized as the boundary of some smooth compact manifold.*

This theorem gives rise to the definition of "cobordism class".

Definition 3.1.26. *Two smooth closed n -manifolds M_1 and M_2 belong to the same **unoriented cobordism class** if and only if their disjoint union $M_1 \sqcup M_2$ is the boundary of a smooth compact $(n+1)$ -manifold.*

Thus we have the following corollary:

Corollary 3.1.27. *Two smooth closed n -manifolds belong to the same cobordism class if and only if all of their corresponding Stiefel-Whitney numbers are equal.*

For more discussion about cobordism, see subsequent sections.

As we pointed out in last section, there is another way to define Stiefel-Whitney classes using universal bundle. Now let's discuss about it.

First we recall the Grassmann manifold:

Definition 3.1.28. *The **Grassmann manifold** $\text{Gr}_n(\mathbb{R}^{n+k})$ is the set of all n -dimensional planes through the origin of the coordinate space $\mathbb{R}^{n+k}$. This is to be topologized as a quotient space, as follows. The collection of all n -frames in $\mathbb{R}^{n+k}$ forms an open set of the n -fold Cartesian product $\mathbb{R}^{n+k} \times \cdots \times \mathbb{R}^{n+k}$, called the **Stiefel manifold** $V_n(\mathbb{R}^{n+k})$. There is a canonical function*

$$q : V_n(\mathbb{R}^{n+k}) \rightarrow \text{Gr}_n(\mathbb{R}^{n+k})$$

which maps each n -frame to the n -plane which it spans. Now give $\text{Gr}_n(\mathbb{R}^{n+k})$ to quotient topology: a subset $U \subseteq \text{Gr}_n(\mathbb{R}^{n+k})$ is open if and only if its inverse image $q^{-1}(U) \subseteq V_n(\mathbb{R}^{n+k})$ is open.

Then we can check that

Lemma 3.1.29. *The Grassmann manifold $\text{Gr}_n(\mathbb{R}^{n+k})$ is a compact topological manifold of dimension nk . The correspondence $X \rightarrow X^\perp$, which assigns to each n -plane its orthogonal k -plane, defines a diffeomorphism between $\text{Gr}_n(\mathbb{R}^{n+k})$ and $\text{Gr}_k(\mathbb{R}^{n+k})$.*

There is a canonical vector bundle $\gamma^n(\mathbb{R}^{n+k})$ over $\text{Gr}_n(\mathbb{R}^{n+k})$ constructed as follows: let $E = E(\gamma^n(\mathbb{R}^{n+k}))$ be the set of all pairs

$$(n\text{-planes in } \mathbb{R}^{n+k}, \text{ vector in that } n\text{-plane})$$

This is topologized as a subset of $\text{Gr}_n(\mathbb{R}^{n+k}) \times \mathbb{R}^{n+k}$. We can check that it's a vector bundle.

If we put all $\text{Gr}_n(\mathbb{R}^{n+k})$ together when fixing n , we get an infinite-dimensional manifold.

Definition 3.1.30. *The **infinite Grassmann manifold***

$$\text{Gr}_n = \text{Gr}_n(\mathbb{R}^\infty)$$

is the set of all n -dimensional linear subspaces of $\mathbb{R}^\infty$, topologized as the direct limit of the sequence

$$\text{Gr}_n(\mathbb{R}^n) \subseteq \text{Gr}_n(\mathbb{R}^{n+1}) \subseteq \text{Gr}_n(\mathbb{R}^{n+2}) \subseteq \dots$$

In other words, a subset of Gr_n is open if and only if its intersection with $\text{Gr}_n(\mathbb{R}^{n+k})$ is open as a subset of $\text{Gr}_n(\mathbb{R}^{n+k})$ for each k . We write

$$\text{Gr}_n = \varinjlim \text{Gr}_n(\mathbb{R}^{n+k})$$

A canonical bundle γ^n over Gr_n is constructed similarly. Let

$$E(\gamma^n) \subseteq \text{Gr}_n \times \mathbb{R}^\infty$$

be the set of all pairs

$$(n\text{-planes in } \mathbb{R}^\infty, \text{ vector in that } n\text{-plane})$$

topologized as a subset of the Cartesian product $\text{Gr}_n \times \mathbb{R}^\infty$. We can also check that it's a vector bundle.

The following two theorems imply that this bundle γ^n over Gr_n is a **universal bundle**.

Theorem 3.1.31. *Any $\mathbb{R}^n$ -bundle ξ over a paracompact basespace admits a bundle map $\xi \rightarrow \gamma^n$.*

Definition 3.1.32. *Two bundle maps $f, g : \xi \rightarrow \gamma^n$ are called **bundle-homotopic** if there exists a one-parameter family of bundle maps*

$$h_t : \xi \rightarrow \gamma^n, \quad 0 \leq t \leq 1$$

with $h_0 = f$, $h_1 = g$, such that h is continuous as a function of both variables.

Theorem 3.1.33. *Any two bundle maps from an $\mathbb{R}^n$ -bundle to γ^n are bundle-homotopic.*

The proof is somehow technical, so we will skip it. As a corollary, we immediately get

Corollary 3.1.34. *Any $\mathbb{R}^n$ -bundle ξ over a paracompact space B determines a unique homotopy class of maps*

$$\bar{f}_\xi : B \longrightarrow \text{Gr}_n$$

Then we can give a precise definition of the concept of characteristic class.

First let Λ be a coefficient group or ring and let

$$c \in H^i(\text{Gr}_n; \Lambda)$$

be any cohomology class. Then ξ and c together determine a cohomology class

$$\bar{f}_\xi^* c \in H^i(B; \Lambda)$$

This class will be denoted by $c(\xi)$.

Definition 3.1.35. *$c(\xi)$ is called the **characteristic cohomology class** of ξ determined by c .*

Note that the correspondence $\xi \rightarrow c(\xi)$ is natural with respect to bundle maps. Conversely, given any correspondence

$$\xi \mapsto c(\xi) \in H^i(B(\xi); \Lambda)$$

which is natural with respect to bundle maps, we have

$$c(\xi) = \bar{f}_\xi^* c(\gamma^n)$$

In conclusion, *the ring consisting of all characteristic cohomology classes for $\mathbb{R}^n$ -bundles over paracompact base spaces with coefficient ring Λ is canonically isomorphic to the cohomology ring $H^*(\text{Gr}_n; \Lambda)$.*

Thus the problem turns into computation for cohomology of Gr_n .

Interlude 2: Cell Structure for Grassmann Manifolds

Before talking about cohomology for Gr_n , we should first know the cell structure so that we can apply cellular homology.

$\mathbb{R}^m$ contains subspaces

$$\mathbb{R}^0 \subseteq \mathbb{R}^1 \subseteq \cdots \subseteq \mathbb{R}^m$$

where $\mathbb{R}^k$ consists of all vectors of the form $v = (v_1, \dots, v_k, 0, \dots, 0)$. Any n -plane $X \subseteq \mathbb{R}^m$ gives rise to a sequence of integers

$$0 \leq \dim(X \cap \mathbb{R}^1) \leq \dim(X \cap \mathbb{R}^2) \leq \cdots \leq \dim(X \cap \mathbb{R}^m) = n$$

Two consecutive integers in this sequence differ by at most 1. Thus the above sequence of integers contains precisely n jumps. We introduce the **Schubert symbol** $\sigma = (\sigma_1, \dots, \sigma_n)$ to represent a sequence of n integers satisfying

$$1 \leq \sigma_1 < \sigma_2 < \cdots < \sigma_n \leq m$$

For each Schubert σ , let $e(\sigma) \subseteq \text{Gr}_n(\mathbb{R}^m)$ denote the set of all n -planes X such that

$$\dim(X \cap \mathbb{R}^{\sigma_i}) = i, \dim(X \cap \mathbb{R}^{\sigma_i-1}) = i-1$$

We can show that

Theorem 3.1.36. *$e(\sigma)$ is open cell of dimension*

$$d(\sigma) = (\sigma_1 - 1) + (\sigma_2 - 2) + \cdots + (\sigma_n - n)$$

The $\binom{m}{n}$ sets $e(\sigma)$ form the cells of a CW-complex with underlying space $\text{Gr}_n(\mathbb{R}^m)$. Similarly taking the direct limit as $m \rightarrow \infty$, one obtains an infinite CW-complex with underlying space $\text{Gr}_n = \text{Gr}_n(\mathbb{R}^\infty)$.

If we want to count the number of r -cells in $\text{Gr}_n(\mathbb{R}^m)$ for arbitrary n , we can introduce the following definition.

Definition 3.1.37. A *partition of an integer* $r \geq 0$ is an unordered sequence $i_1 \cdots i_s$ of positive integers with sum r . The number of partitions of r is denoted by $p(r)$.

To every Schubert symbol $(\sigma_1, \dots, \sigma_n)$ with $d(\sigma) = r$ and $\sigma_n \leq m$ there corresponds a partition $i_1 \cdots i_s$ of r , where $i_1, \dots, i_s$ denotes the sequence obtained from $\sigma - 1, \dots, \sigma_n - n$ by canceling any zeros appearing. Thus

Corollary 3.1.38. The number of r -cells in $\text{Gr}_n(\mathbb{R}^m)$ is equal to the number of partitions of r into at most n integers each of which is $\leq m - n$.

Note that this corollary is also true for $m = +\infty$.

Now let's come back to calculate the cohomology ring of Gr_n . First we give the conclusion.

Theorem 3.1.39. The cohomology ring $H^*(\text{Gr}_n; \mathbb{Z}_2)$ is a polynomial algebra over $\mathbb{Z}_2$ freely generated by the Stiefel-Whitney classes $w_1(\gamma^n), \dots, w_n(\gamma^n)$, where γ^n is the universal bundle over Gr_n .

To prove this, we first need show the following.

Lemma 3.1.40. There are no polynomial relations among the $w_i(\gamma^n)$.

Proof. Suppose there is a relation $p(w_1(\gamma^n), \dots, w_n(\gamma^n)) = 0$. By theorem 3.1.31, for any n -plane bundle ξ over a paracompact base space there exists a bundle map $g : \xi \rightarrow \gamma^n$. Hence

$$w_i(\xi) = \bar{g}^*(w_i(\gamma^n))$$

where $\bar{g}$ is the map of basespaces induced by g . It follows that $w_i(\xi)$ also satisfy the corresponding relation

$$p(w_1(\xi), \dots, w_n(\xi)) = \bar{g}^*p(w_1(\gamma^n), \dots, w_n(\gamma^n)) = 0$$

Thus we only need to find some n -plane bundle ξ so that there are no polynomial relations among the classes $w_1(\xi), \dots, w_n(\xi)$.

Consider the canonical line bundle γ^1 over $\mathbb{RP}^\infty$. Recall that $H^*(\mathbb{RP}^\infty; \mathbb{Z}_2)$ is a polynomial algebra over $\mathbb{Z}_2$ with a single generator a of dimension 1, and $w(\gamma^1) = 1 + a$. Let X be the n -fold Cartesian product $X = \mathbb{RP}^\infty \times \cdots \times \mathbb{RP}^\infty$, by Künneth formula, $H^*(X; \mathbb{Z}_2)$ is a polynomial algebra on n generators $a_1, \dots, a_n$ of dimension 1. Let ξ be the n -fold Cartesian product

$$\xi = \gamma^1 \times \cdots \times \gamma^1 \cong (\pi_1^* \gamma^1) \oplus \cdots \oplus (\pi_n^* \gamma^1)$$

Then ξ is an n -plane bundle over $X = \mathbb{RP}^\infty \times \cdots \times \mathbb{RP}^\infty$, and the total Stiefel Whitney class

$$w(\xi) = w(\gamma^1) \times \cdots \times w(\gamma^1) = (1 + a_1)(1 + a_2) \cdots (1 + a_n)$$

In other words, $w_k(\xi)$ is the k -th elementary symmetric function of $a_1, \dots, a_n$. And we know in algebra that the n elementary symmetric functions in n indeterminates over a field are algebraically independent (see [23]). Hence we have done. $\square$

Now we prove theorem 3.1.39.

Proof. We have shown that $H^*(\text{Gr}_n; \mathbb{Z}_2)$ contains a polynomial algebra over $\mathbb{Z}_2$ freely generated by $w_1(\gamma^n), \dots, w_n(\gamma^n)$. Using a counting argument, we will show that this subalgebra actually coincides with $H^*(\text{Gr}_n; \mathbb{Z}_2)$.

By corollary 3.1.38, the number of r -cells in Gr_n is equal to the number of partitions of r into at most n integers. Hence the rank of $H^r(\text{Gr}_n; \mathbb{Z}_2)$ is at most equal to this number of partitions. On the other hand the number of distinct monomials of the form

$$w_1(\gamma^n)^{r_1} \dots w_n(\gamma^n)^{r_n}$$

in $H^r(\text{Gr}_n; \mathbb{Z}_2)$ is also precisely equal to the number of partitions of r into at most n integers. This is because, for each sequence $e_1, \dots, r_n$ of non-negative integers with

$$r_1 + 2r_2 + \dots + nr_n = r$$

we can associate the partition of r which is obtained from the n -tuple

$$r_n, r_n + r_{n-1}, \dots, r_n + r_{n-1} + \dots + r_1$$

by deleting any zeros which may occur, and vice versa.

Since these monomials are known to be linearly independent mod 2, it follows that the inequalities above must all actually be equalities. Hence $H^r(\text{Gr}_n)$ has a basis consisting of the monomials $w_1(\gamma^n)^{r_1} \dots w_n(\gamma^n)^{r_n}$ of total dimension r . $\square$

Now we can prove the uniqueness of Stiefel-Whitney classes.

Theorem 3.1.41 (Uniqueness Theorem). *There only exists one correspondence $\xi \rightarrow w(\xi)$ which assigns to each vector bundle over a paracompact base space a sequence of cohomology classes satisfying the four axioms for Stiefel-Whitney classes.*

Proof. Suppose that there are two, say $\xi \rightarrow w(\xi)$ and $\xi \rightarrow \tilde{w}(\xi)$. For the canonical line bundle γ_1^1 over $\mathbb{RP}^1$ we have

$$w(\gamma_1^1) = \tilde{w}(\gamma_1^1) = 1 + a$$

by Axiom 1 and 4. Embedding γ_1^1 in the line bundle γ^1 over $\mathbb{RP}^\infty$, it follows that

$$w(\gamma^1) = \tilde{w}(\gamma^1) = 1 + a$$

by Axiom 1 and 2. Passing to the n -fold Cartesian product

$$\xi = \gamma^1 \times \dots \times \gamma^1 \cong \pi_1^* \gamma^1 \oplus \dots \oplus \pi_n^* \gamma^1$$

it follows that

$$w(\xi) = \tilde{w}(\xi) = (1 + a_1) \dots (1 + a_n)$$

by Axiom 2 and 3. Now using the existence of a bundle map $\xi \rightarrow \gamma^n$, and the fact that $H^*(\text{Gr}_n)$ injects monomorphically into $H^*(\mathbb{RP}^\infty \times \dots \times \mathbb{RP}^\infty)$ it follows that $w(\gamma^n) = \tilde{w}(\gamma^n)$.

For any n -plane bundle η over a paracompact base space, choosing a bundle map $f : \eta \rightarrow \gamma^n$, it follows that

$$w(\eta) = \bar{f}^* w(\gamma^n) = \bar{f}^* \tilde{w}(\gamma^n) = \tilde{w}(\eta)$$

$\square$

Our discussion on Stiefel-Whitney classes has come to a temporary end. In conclusion, we propose an axiom system for Stiefel-Whitney classes and prove its existence and uniqueness. Besides, we establish the classifying space Grassmann manifolds and the universal bundle. Using cohomology of Grassmann manifolds, we can obtain Stiefel-Whitney classes on any vector bundle by pullback.

3.2. Euler class.

Now we turn to another characteristic class defined on oriented bundle with coefficient $\mathbb{Z}$, called the Euler class.

We first recall the definition of oriented bundle.

Definition 3.2.1. An **orientation** of a vector bundle ξ is a function which assigns an orientation to each fiber F of ξ , subject to the following local compatibility condition. For every point $b_0 \in B$ there exists a local coordinate (N, h) with $b_0 \in N$ and $h : N \times \mathbb{R}^n \rightarrow \pi^{-1}(N)$, so that for each fiber $F = \pi^{-1}(b)$ over N the homomorphism $x \mapsto h(b, x)$ from $\mathbb{R}^n$ to F is orientation preserving.

As we discussed before, we have Thom isomorphism for oriented bundles.

Theorem 3.2.2 (Thom isomorphism, see [13]). Let ξ be an oriented n -plane bundle with total space E . Then the cohomology group $H^i(E, E_0; \mathbb{Z})$ is zero for $i < n$, and $H^n(E, E_0; \mathbb{Z})$ contains one and only one cohomology class u whose restriction

$$u|_{(F, F_0)} \in H^n(F, F_0; \mathbb{Z})$$

is equal to the preferred generator u_F for every fiber F of ξ . Furthermore the correspondence $y \mapsto y \smile u$ maps $H^k(E, \mathbb{Z})$ isomorphically onto $H^{k+n}(E, E_0; \mathbb{Z})$ for every integer k .

Definition 3.2.3. The **Thom isomorphism** ϕ is defined to be the composition of the two isomorphisms

$$H^k(B; \mathbb{Z}) \xrightarrow{\pi^*} H^k(E; \mathbb{Z}) \xrightarrow{\smile u} H^{k+n}(E, E_0; \mathbb{Z})$$

Now we can define an important new characteristic class. Given an oriented n -plane bundle ξ , we have a restriction homomorphism

$$H^*(E, E_0; \mathbb{Z}) \rightarrow H^*(E; \mathbb{Z})$$

which we denote by $y \mapsto y|_E$. In particular, applying this homomorphism to the fundamental class $u \in H^n(E, E_0; \mathbb{Z})$, we obtain a new cohomology class

$$u|_E \in H^n(E; \mathbb{Z})$$

But $H^n(E; \mathbb{Z})$ is canonically isomorphic to $H^n(B; \mathbb{Z})$.

Definition 3.2.4. The **Euler class** of an oriented n -plane bundle ξ is the cohomology class

$$e(\xi) \in H^n(B; \mathbb{Z})$$

which corresponds to $u|_E$ under the canonical isomorphism $\pi^* : H^n(B; \mathbb{Z}) \rightarrow H^n(E; \mathbb{Z})$.

We have some fundamental properties of the Euler class:

Proposition 3.2.5 (Naturality). If $f : B \rightarrow B'$ is covered by an orientation preserving bundle map $\xi \rightarrow \xi'$, then $e(\xi) = f^*e(\xi')$.

In particular, if ξ is a trivial n -plane bundle, then $e(\xi) = 0$ since we can take ξ' to be a bundle over a point.

Proposition 3.2.6. *If the orientation of ξ is reversed, then the Euler class $e(\xi)$ changes sign.*

Proof. The orientation of each fiber is reversed, hence u changes sign. $\square$

Proposition 3.2.7. *If the fiber dimension n is odd, then $e(\xi) + e(\xi) = 0$.*

We can use last proposition to prove this, but here we give an alternative proof.

Proof. The Thom isomorphism $\phi(x) = \pi^*(x) \smile u$ maps $e(\xi)$ to

$$\pi^*e(\xi) \smile u = (u|_E) \smile u = u \smile u$$

Hence

$$e(\xi) = \phi^{-1}(u \smile u)$$

Since n is odd, $u \smile u$ is an element of order 2. $\square$

Note that in the proof, we give an explicit formula for Euler class.

Proposition 3.2.8. *The natural homomorphism $H^n(B; \mathbb{Z}) \rightarrow H^n(B; \mathbb{Z}_2)$ carries the Euler class $e(\xi)$ to the top Stiefel-Whitney class $w_n(\xi)$.*

Proof. Under $\mathbb{Z} \rightarrow \mathbb{Z}_2$, the two fundamental class u are coincide. Note that $Sq^n(u) = u \smile u$. Hence $e(\xi) = \phi^{-1}(u \smile u)$ maps to $\phi^{-1}Sq^n(u) = w_n(\xi)$. $\square$

There are several properties of $w_n(\xi)$ applying equally well to $e(\xi)$.

Proposition 3.2.9. *The Euler class of a Whitney sum is given by $e(\xi \oplus \xi') = e(\xi) \smile e(\xi')$. Similarly the Euler class of a Cartesian product is given by $e(\xi \times \xi') = e(\xi) \times e(\xi')$.*

Proof. As calculation in construction for Stiefel-Whitney class, we have

$$u(\xi \times \xi') = (-1)^{mn} u(\xi) \times u(\xi')$$

for two oriented bundles ξ, ξ' with fiber dimensions m and n . Now apply π^* , we get

$$e(\xi \times \xi') = (-1)^{mn} e(\xi) \times e(\xi')$$

where the sign can be ignored since the right side of this equation is an element of order two whenever m or n is odd. $\square$

Proposition 3.2.10. *If the oriented vector bundle ξ possesses a nowhere zero cross-section, then the Euler class $e(\xi)$ must be zero.*

Proof. Let $s : B \rightarrow E_0$ be a cross-section, then the composition

$$B \xrightarrow{s} E_0 \subseteq E \xrightarrow{\pi} B$$

is the identity, thus it induces identity on $H^n(B)$. But the Euler class $\pi^*(u|_E)$ factors through $H^n(E_0)$, which is zero since $u \in H^n(E, E_0)$. Hence $e(\xi) = 0$. $\square$

3.3. Application on smooth manifolds.

Let $M = M^n$ be a smooth manifold which embedded in a Riemannian manifold $A = A^{n+k}$, we look at the normal bundle of M in A , and let's see what we can say about its characteristic classes.

Recall that M has a tubular neighborhood in A which can be given by the exponential map: If we let E be the total space of the normal bundle ν^k , $E(\varepsilon)$ be a small neighborhood of zero section in E , then the exponential map

$$\exp : E(\varepsilon) \longrightarrow A$$

maps $E(\varepsilon)$ diffeomorphically onto a neighborhood N_ε of M in A , see [24].

Proposition 3.3.1. *If M is closed in A , then the cohomology ring $H^*(E, E_0; \Lambda)$ associated with the normal bundle of M in A is canonically isomorphic to the cohomology ring $H^*(A, A - M; \Lambda)$ for any ring Λ .*

Proof. Since N_ε and $A - M$ are open subsets with union A and intersection $N_\varepsilon - M$, there is an excision isomorphism

$$H^*(A, A - M) \longrightarrow H^*(N_\varepsilon, N_\varepsilon - M)$$

Therefore the embedding

$$\exp : (E(\varepsilon), E(\varepsilon)_0) \longrightarrow (N_\varepsilon, N_\varepsilon - M) \subseteq (A, A - M)$$

induces an isomorphism

$$\exp^* : H^*(A, A - M) \longrightarrow H^*(E(\varepsilon), E(\varepsilon)_0)$$

Composing with the excision isomorphism

$$H^*(E(\varepsilon), E(\varepsilon)_0) \cong H^*(E, E_0)$$

we obtain the required isomorphism. $\square$

As a direct corollary, the fundamental cohomology class $u \in H^k(E, E_0; \mathbb{Z}_2)$ corresponds to a canonical class which we denote by

$$u' \in H^k(A, A - M; \mathbb{Z}_2)$$

Similarly, if ν_k is orientable, then any orientation for ν_k determines a corresponding class $u' \in H^k(A, A - M; \mathbb{Z})$.

Theorem 3.3.2. *If M is embedded as a closed subset of A , then the composition of the two restriction homomorphisms*

$$H^k(A, A - M) \longrightarrow H^k(A) \longrightarrow H^k(M)$$

with mod 2 coefficients, maps the fundamental class u' to the top Stiefel-Whitney class $w_k(\nu^k)$ of the normal bundle. Similarly, if ν^k is oriented, then the corresponding composition with integer coefficients maps the integral fundamental class u' to the Euler class $e(\nu^k)$.

Proof. Let $s : M \longrightarrow E$ denote the zero section of ν^k , inducing a canonical isomorphism $H^*(E) \longrightarrow H^*(M)$. First note that the composition

$$H^k(E, E_0) \longrightarrow H^k(E) \xrightarrow{s^*} H^k(M)$$

with mod 2 coefficients maps the fundamental class u to the Stiefel-Whitney class $w_k(\nu^k)$. In fact the image of $s^*(u|_E)$ under the Thom isomorphism

$$\phi : H^k(M) \rightarrow H^{2k}(E, E_0)$$

is equal to

$$\pi^* s^*(u|_E) \smile u = (u|_E) \smile u = u \smile u = Sq^k(u)$$

Therefore $s^*(u|_E)$ is equal to $\phi^{-1}(Sq^k(u)) = w_k(\nu^k)$.

Now by the previous proposition, we get the theorem. $\square$

Definition 3.3.3. *The image of u' in $H^k(A)$ is called the **dual cohomology class to the submanifold M of codimension k** .*

If this dual class $u'|_A$ is zero, then the top Stiefel-Whitney class (or the Euler class) of ν^k must be zero. One special case is particularly useful:

Corollary 3.3.4. *If $M = M^n$ is smoothly embedded as a closed subset of the Euclidean space $\mathbb{R}^{n+k}$, then $w_k(\nu^k) = 0$. In the oriented case $e(\nu^k) = 0$.*

For example, if n is a power of 2, then $\mathbb{R}P^n$ cannot be smoothly embedded in $\mathbb{R}^{2n-1}$. Hence it gives a lower bound on strong Whitney embedding theorem.

Now we consider M a Riemannian manifold. Then the product $M \times M$ also has the structure of a Riemannian manifold. We have the following lemmas:

Lemma 3.3.5. *The normal bundle ν^n associated with the diagonal embedding of M in $M \times M$ is canonically isomorphic to the tangent bundle of M .*

Lemma 3.3.6. *Any orientation for the tangent bundle τ_M gives rise to an orientation for the underlying topological manifold M , and conversely any orientation for M gives rise to an orientation for τ_M .*

Proof. Applying proposition 3.3.1 to the 0-dimensional manifold x in M with normal bundle $T_x M$, we know that $H_n(M, M - x)$ is canonically isomorphic to $H_n(T_x M, T_x M - 0)$. Then by the definition of orientation, we get the result. $\square$

Now we fix a ring Λ and study homology and cohomology of M . We will always assume either M is oriented or $\Lambda = \mathbb{Z}_2$. It follows from proposition 3.3.1 that there is a fundamental cohomology class

$$u' \in H^n(M \times M, M \times M - \Delta(M))$$

with coefficient in Λ . By theorem 3.3.2, the restriction of u' to the diagonal submanifold $\Delta(M) \cong M$ is equal to the Euler class

$$e(\nu^n) = e(\tau_M)$$

with coefficient ring Λ , in the oriented case, or to the Stiefel-Whitney class $w_n(\tau_M)$ in the mod 2 case.

u' can be characterized as follows. Note that each $H^n(M, M - x)$ has a preferred generator u_x , defined by the condition

$$\langle u_x, \mu_x \rangle = 1$$

Define the canonical embedding

$$j_x : (M, M - x) \longrightarrow (M \times M, M \times M - \Delta(M))$$

by setting $j_x(y) = (x, y)$.

Lemma 3.3.7. *The class $u' \in H^n(M \times M, M \times M - \Delta(M))$ is uniquely characterized by the property that its image $j_x^*(u')$ is equal to the preferred generator u_x for every $x \in M$.*

Proof. For any x and any small neighborhood N of zero in $T_x M$, consider the embedding

$$(N, N - 0) \longrightarrow (M \times M, M \times M - \Delta(M))$$

defined by the exponential map

$$v \longmapsto (\exp(x, -v), \exp(x, v))$$

Then the induced cohomology homomorphism must map u' to the preferred generator for $H^n(N, N - 0) \cong H^n(T_x M, T_x M - 0)$. Using the homotopy $(v, t) \mapsto (\exp(x, -tv), \exp(x, v))$ for $0 \leq t \leq 1$, the previous map is homotopic to the map

$$v \mapsto (x, \exp(x, v))$$

This is the composition of j_x with the canonical embedding

$$\exp : (N, N - 0) \longrightarrow (M, M - x)$$

the conclusion follows. $\square$

Now consider the restriction homomorphism

$$H^n(M \times M, M \times M - \Delta(M)) \longrightarrow H^n(M \times M)$$

maps u' to a cohomology class $u'|_{M \times M}$ which, by definition, is “dual” to the diagonal submanifold of $M \times M$.

Definition 3.3.8. *This cohomology class $u'|_{M \times M}$ will be denoted by u'' , and called the **diagonal cohomology class** in $H^n(M \times M)$.*

Lemma 3.3.9. *For any cohomology class $a \in H^*(M)$, the product $(a \times 1) \smile u''$ is equal to $(1 \times a) \smile u''$.*

Proof. Let N_ε be a tubular neighborhood of $\Delta(M)$ in $M \times M$, then $\Delta(M)$ is a deformation retract of N_ε . Define two projection maps

$$\pi_1, \pi_2 : M \times M \rightarrow M$$

Since π_1 and π_2 coincide on $\Delta(M)$, it follows that the restriction $\pi_1|_{N_\varepsilon}$ is homotopic to $\pi_2|_{N_\varepsilon}$. Therefore the two cohomology classes $\pi_1^*(a) = a \times 1$ and $\pi_2^*(a) = 1 \times a$ have the same image under the restriction homomorphism $H^i(M \times M) \rightarrow H^i(N_\varepsilon)$. Now using the commutative diagram

$$\begin{array}{ccc} H^i(M \times M) & \xrightarrow{\quad\quad\quad} & H^i(N_\varepsilon) \\ \downarrow \smile u' & & \downarrow \smile u'|_{N_\varepsilon, N_\varepsilon - \Delta(M)} \\ H^{i+n}(M \times M, M \times M - \Delta(M)) & \xrightarrow{\cong} & H^{i+n}(N_\varepsilon, N_\varepsilon - \Delta(M)) \end{array}$$

it follows that $(a \times 1) \smile u' = (1 \times a) \smile u'$. Restricting to $H^{i+n}(M \times M)$, the conclusion follows. $\square$

Now we define a new operation called **slant product** as follows. Let X and Y are finite CW complexes and Λ a field, define a homomorphism

$$H^*(X \times Y) \otimes H_*(Y) = H^*(X) \otimes H^*(Y) \otimes H_*(Y) \rightarrow H^*(X)$$

by the formula $p \otimes \beta = a \otimes b \otimes \beta \mapsto a \langle b, \beta \rangle$, which is denoted by p/β . This operation satisfies and is characterized by the identity

$$(a \times b)/\beta = a \langle b, \beta \rangle$$

For each fixed $\beta \in H_*(Y)$, note that the homomorphism $p \mapsto p/\beta$ is left $H^*(X)$ -linear in the sense that $((a \times 1) \smile p)/\beta = a \smile (p/\beta)$ for every $a \in H^*(X)$ and $p \in H^*(X \times Y)$. For general X, Y and Λ , we can also define slant product.

Lemma 3.3.10. *Suppose M is compact, so that the fundamental class $\mu \in H_n(M)$ is defined. Then the diagonal cohomology class $u'' \in H^n(M \times M)$ and the fundamental class μ are related by the identity $u''/\mu = 1 \in H^0(M)$.*

Proof. For any $x \in M$, we will compute the image of u''/μ under the restriction homomorphism $H^0(M) \rightarrow H^0(x) \cong \Lambda$. We will make use of the commutative diagram

$$\begin{array}{ccc} H^n(M \times M) & \xrightarrow{/\mu} & H^0(M) \\ \downarrow & & \downarrow \\ H^n(x \times M) & \xrightarrow{/\mu} & H^0(x) \end{array}$$

Note that the left hand vertical arrow maps u'' to $1 \times i_x^*(u'')$, where

$$i_x : M \rightarrow M$$

denotes the embedding $y \mapsto (x, y)$. Using the identity $(a \times b)/\mu = a\langle b, \mu \rangle$, it follows that $(u''/\mu)|_x$ is equal to the Kronecker index $\langle i_x^*(u''), \mu \rangle$ multiplied by $1 \in H^0(x)$.

Note that, for each $x \in M$, the natural homomorphism

$$H_n(M) \rightarrow H_n(M, M - x)$$

maps μ to the preferred generator μ_x . Making use of the mappings

$$\begin{array}{ccc} M & \longrightarrow & (M, M - x) \\ \downarrow i_x & & \downarrow j_x \\ M \times M & \longrightarrow & (M \times M, M \times M - \Delta(M)) \end{array}$$

where j_x also sends y to (x, y) , it follows that the Kronecker index $\langle i_x^*(u''), \mu \rangle = \langle j_x^*(u'')|_M, \mu \rangle$ is equal to $\langle j_x^*(u''), \mu_x \rangle$. Since this equals 1 by lemma 3.3.7, we have proved that

$$(u''/\mu)|_x = 1 \in H^0(x)$$

Hence u''/μ is equal to the identity element of $H^0(M)$. $\square$

Now consider the following form of duality theorem:

Theorem 3.3.11. *To each basis $b_1, \dots, b_r$ for $H^*(M)$ there corresponds a dual basis $b_1^\#, \dots, b_r^\#$ for $H^*(M)$, satisfying the identity*

$$\langle b_i \smile b_j^\#, \mu \rangle = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Theorem 3.3.12. *With $\{b_i\}$ and $\{b_i^\#\}$ as above, the diagonal cohomology class u'' is equal to*

$$\sum_{i=1}^r (-1)^{\dim b_i} b_i \times b_i^\#$$

We will prove these two theorems simultaneously

Proof. Using the Künneth formula, we can write u'' as

$$u'' = b_1 \times c_1 + \cdots + b_r \times c_r$$

where $c_1, \dots, c_r$ are certain well-defined cohomology classes in $H^*(M)$ with

$$\dim b_i + \dim c_i = n$$

Apply the homomorphism μ to both sides of the identity

$$(a \times 1) \smile u'' = (1 \times a) \smile u''$$

On the left side, we obtain

$$((a \times 1) \smile u'')/\mu = a \smile (u''/\mu) = a$$

On the right side, we obtain

$$\sum (-1)^{\dim a \dim b_j} (b_j \times (a \smile c_j))/\mu = \sum (-1)^{\dim a \dim b_j} b_j \langle a \smile c, \mu \rangle$$

Substituting b_i for a , it follows that the coefficient

$$(-1)^{\dim a \dim b_j} \langle b_i \smile c_j, \mu \rangle$$

of b_j must be 1 for $i = j$, and 0 for $i \neq j$. Setting $b_i^\# = (-1)^{\dim b_i} c_i$, the conclusion follows easily. $\square$

Now we have the most important corollary, which explains the name of Euler class:

Theorem 3.3.13. *If M is a smooth compact oriented manifold with fundamental class μ , then the Kronecker index $\langle e(\tau_M), \mu \rangle$, using rational or integer coefficients, is equal to the Euler characteristic $\chi(M)$. Similarly, for a non-oriented manifold, the Stiefel-Whitney number $\langle w_n(\tau_M), \mu \rangle = w_n[M]$ is congruent to $\chi(M)$ modulo 2.*

Proof. By theorem 3.3.2, the Euler class of the tangent bundle is given by

$$e(\tau_M) = \Delta^*(u'')$$

Using rational coefficients, We can substitute the formula

$$u'' = \sum (-1)^{\dim b_i} b_i \times b_i^\#$$

thus obtain

$$e(\tau_M) = \sum (-1)^{\dim b_i} b_i \smile b_i^\#$$

Now applying the homomorphism $\langle -, \mu \rangle$ to both sides. we obtain

$$\langle e(\tau_M), \mu \rangle = \sum (-1)^{\dim b_i} = \chi(M)$$

$\square$

3.4. Obstruction theory.

As we mentioned before, characteristic classes were defined as obstructions to the existence of independent vector fields in the original work of Stiefel and Whitney. Besides, vector bundles admit more good structures, hence we can say more about it than the classical obstruction theory. Now we discuss it in detail. (The reader should know some basic obstruction theory, see [21].)

Let ξ be an n -plane bundle with base space B . For each fiber F of ξ , consider the Stiefel manifold $V_k(F)$ consisting of all k -frames in F . These manifolds $V_k(F)$ can be considered as the fibers of a new fiber bundle which we denote by $V_k(\xi)$ and call it the **associated Stiefel manifold bundle** over B . Note that a section of this Stiefel manifold bundle is a k -tuple of linearly independent sections of ξ .

Assume B is a CW complex, Steenrod shows that the fiber $V_k(F)$ is $(n-k-1)$ -connected, so we can construct a section of $V_k(\xi)$ over the $(n-k)$ -skeleton of B by classical obstruction theory. There exists a section over the $(n-k+1)$ -skeleton of B if and only if the **primary obstruction class** in

$$H^{n-k+1}(B; \pi_{n-k}(V_k(F)))$$

is zero. Setting $j = n - k + 1$, we will use the notation

$$\mathfrak{o}_j(\xi) \in H^j(B; \pi_{j-1}(V_{n-j+1}(F)))$$

for this primary obstruction class. If j is even, and less than n , then Steenrod shows that the coefficient group $\pi_{j-1}(V_{n-j+1}(F)) \cong \mathbb{Z}_2$. If j is odd, or $j = n$, $\pi_{j-1}(V_{n-j+1}(F)) \cong \mathbb{Z}$. In any case, there is a unique non-trivial homomorphism h from $\pi_{j-1}(V_{n-j+1}(F))$ to $\mathbb{Z}_2$. Hence we can obtain $h_*\mathfrak{o}_j(\xi) \in H^j(B; \mathbb{Z}_2)$.

Theorem 3.4.1. *$h_*\mathfrak{o}_j(\xi)$ is equal to the Stiefel-Whitney class $w_j(\xi)$.*

Proof. First consider the universal bundle γ^n over $\text{Gr}_n = \text{Gr}_n(\mathbb{R}^\infty)$. Since $H^*(\text{Gr}_n; \mathbb{Z}_2)$ is a polynomial algebra generated by $w_1(\gamma^n), \dots, w_n(\gamma^n)$, it follows that

$$h_*\mathfrak{o}_j(\gamma^n) = f_j(w_1(\gamma^n), \dots, w_n(\gamma^n))$$

for some polynomial f_j in n variables. By naturality,

$$h_*\mathfrak{o}_j(\xi) = f_j(w_1(\xi), \dots, w_n(\xi))$$

for any n -plane bundle ξ .

Since f_j is of dimension $j \leq n$, f_j can be written uniquely as a sum

$$f_j(w_1, \dots, w_n) = f'(w_1, \dots, w_{j-1}) + \lambda w_j$$

where $f' = f'_{j,n}$ is a polynomial and $\lambda = \lambda_{j,n}$ equals to 0 or 1.

To compute f' , consider the n -plane bundle $\eta = \gamma^{j-1} \oplus \varepsilon^{n-j+1}$ over Gr_{n-1} . This bundle η admits $n-j+1$ linearly independent sections, so the obstruction class

$$\mathfrak{o}_j(\eta) \in H^j(B; \pi_{j-1}(V_{n-j+1}(F)))$$

must be zero. Therefore the mod 2 class

$$h_*(\mathfrak{o}_j(\eta)) = f'(w_1(\eta), \dots, w_{j-1}(\eta)) + \lambda w_j(\eta) = f'(w_1(\gamma^{j-1}), \dots, w_{j-1}(\gamma^{j-1})) + 0$$

is equal to zero. Since $w_1(\gamma^{j-1}), \dots, w_{j-1}(\gamma^{j-1})$ are algebraically independent, this proves that $f' = 0$. Thus

$$h_*(\mathfrak{o}_j(\xi)) = \lambda w_j(\xi)$$

for any n -plane ξ .

To prove that $\lambda = \lambda_{j,n} = 1$, first consider the case $j = n$. Let $\xi = \gamma_1^n$ be

the restriction of the universal bundle γ^n to $\text{Gr}_n(\mathbb{R}^{n+1})$. Identifying $\text{Gr}_n(\mathbb{R}^{n+1})$ with $\mathbb{RP}^n$, this bundle can be described as follows. Corresponding to each pair of antipodal points $\{u, -u\}$ on S^n one associates the fiber consisting of all vectors v in $\mathbb{R}^{n+1}$ with $u \cdot v = 0$.

A section of γ_1^n which is non-zero except at a single point $\{u_0, -u_0\}$ of $\mathbb{RP}^n$ is given by the formula $\{u, -u\} \mapsto u_0 - (u_0 \cdot u)u$.

Now choosing u_0 in the middle of the n -dimensional cell of $\mathbb{RP}^n$, we have a section of $V_1(\gamma_1^n)$ over the $(n-1)$ -skeleton, and the obstruction cocycle clearly assigns to the n -cell a generator of the cyclic group

$$\pi_{n-1}(V_1(F)) = \pi_{n-1}(F - 0) \cong \mathbb{Z}$$

Thus $h_* \mathfrak{o}_n(\gamma_1^n) \neq 0$, so $\lambda_{n,n} = 1$.

The proof for $j < n$ is completely analogous. $\square$

Now the question becomes: if we are given the Stiefel-Whitney classes $w_j(\xi)$ of an n -plane bundle, to what extent is it possible to reconstruct the obstruction classes $\mathfrak{o}_j(\xi)$? If $j = 2i$ is even and less than n , then

$$\mathfrak{o}_{2i}(\xi) = w_{2i}(\xi)$$

Next, we will show that if ξ is oriented, the highest obstruction class $\mathfrak{o}_n(\xi)$ can be identified with the Euler class $e(\xi)$. We need a tool called the Gysin sequence.

Theorem 3.4.2. *To any oriented n -plane bundle ξ there is associated an exact sequence of the form*

$$\cdots \rightarrow H^i(B) \xrightarrow{\smile_e} H^{i+n}(B) \xrightarrow{\pi_0^*} H^{i+n}(E_0) \rightarrow H^{i+1}(B) \xrightarrow{\smile_e} \cdots$$

using integer coefficients.

Proof. Starting with the cohomology exact sequence

$$\cdots \rightarrow H^j(E, E_0) \rightarrow H^j(E) \rightarrow H^j(E_0) \xrightarrow{\delta} H^{j+1}(E, E_0) \rightarrow \cdots$$

of the pair (E, E_0) , use the isomorphism

$$\smile u : H^{j-n}(E) \longrightarrow H^j(E, E_0)$$

to substitute $H^{j-n}(E)$ in place of $H^j(E, E_0)$. Thus we obtain an exact sequence of the form

$$\cdots \rightarrow H^{j-n}(E) \xrightarrow{g} H^j(E) \rightarrow H^j(E_0) \rightarrow H^{j-n+1}(E) \rightarrow \cdots$$

where

$$g(x) = (x \smile u)|_E = x \smile (u|_E)$$

Now substitute $H^*(B)$ in place of $H^*(E)$. Since $u|_E$ in $H^n(E)$ corresponds to the Euler class in $H^n(B)$, this yields the sequence we want. $\square$

Similarly, for an unoriented bundle, there is also a corresponding exact sequence with mod 2 coefficients, using the Stiefel-Whitney class $w_n(\xi)$ in place of the Euler class. This can be particularly applied to the following case:

Consider a 2-fold covering $\tilde{B} \rightarrow B$. Then we can construct a line bundle ξ over B whose total space E is obtained from $\tilde{B} \times \mathbb{R}$ by identifying each pair (x, t) with $(x', -t)$ where x and x' are the two distinct points of $\tilde{B}$ lying over one point of B . Then the open subset E_0 contains $\tilde{B}$ as deformation retract. Thus we have proved the following:

Corollary 3.4.3. *To any 2-fold $\tilde{B} \rightarrow B$ there is associated an exact sequence of the form*

$$\cdots \rightarrow H^{j-1}(B) \xrightarrow{\sim w_1} H^j(B) \rightarrow H^j(\tilde{B}) \rightarrow H^j(B) \rightarrow \cdots$$

with mod 2 coefficients.

In contrast with Gr , we now consider the Grassmann manifold consisting of all oriented n planes in $(n+k)$ -space, which we denote by $\widetilde{\text{Gr}}_n(\mathbb{R}^{n+k})$. Clearly $\widetilde{\text{Gr}}_n(\mathbb{R}^{n+k})$ is a 2-fold covering space of the unoriented Grassmann manifold $\text{Gr}_n(\mathbb{R}^{n+k})$. Passing to the direct limit as $k \rightarrow \infty$, we obtain an infinite CW complex $\widetilde{\text{Gr}}_n = \widetilde{\text{Gr}}_n(\mathbb{R}^\infty)$.

Remark 3.4.4. *We also use $\text{BSO}(n)$ to denote the classifying space $\widetilde{\text{Gr}}_n$.*

The universal bundle γ^n over Gr_n lifts to an **oriented universal bundle** n -plane bundle over $\widetilde{\text{Gr}}_n$, which we denote by $\tilde{\gamma}^n$. Then for any oriented n -plane bundle ξ , each bundle map $\xi \rightarrow \gamma^n$ lifts uniquely to an orientation preserving bundle map $\xi \rightarrow \tilde{\gamma}^n$.

Theorem 3.4.5. *The cohomology $H^*(\widetilde{\text{Gr}}_n; \mathbb{Z}_2)$ is a polynomial algebra over $\mathbb{Z}_2$, freely generated by the Stiefel-Whitney classes $w_2(\tilde{\gamma}^n), \dots, w_n(\tilde{\gamma}^n)$.*

Proof. By corollary 3.4.3, there is an exact sequence

$$\cdots \rightarrow H^{j-1}(\text{Gr}_n) \xrightarrow{\sim c} H^j(\text{Gr}_n) \xrightarrow{p^*} H^j(\widetilde{\text{Gr}}_n) \rightarrow \cdots$$

where c is the first Stiefel-Whitney class of the line bundle associated with the 2-fold covering, and $p : \widetilde{\text{Gr}}_n \rightarrow \text{Gr}_n$ the natural map. c cannot be zero, for otherwise, $H^0(\widetilde{\text{Gr}}_n) \cong \mathbb{Z} \oplus \mathbb{Z}$, contradicting with the fact that $\widetilde{\text{Gr}}_n$ is connected. Hence $c = w_1(\gamma^n)$ by theorem 3.1.39. Then the theorem follows. $\square$

In particular, the theorem implies that $H^1(\widetilde{\text{Gr}}_n; \mathbb{Z}_2) = 0$. It follows that $w_1(\widetilde{\text{Gr}}_n) = 0$, and hence that $w_1(\xi) = 0$ for any oriented vector bundle ξ over a paracompact base space.

Remark 3.4.6. *This also tells us the principle that if we want to study a general property, we only need to study this property for the "universal" space. In vector bundle, the "universal" space is Gr_n and $\widetilde{\text{Gr}}_n$. In homotopy theory, the "universal" space is $K(G, n)$, etc.*

No let's study the top obstruction class

$$\mathfrak{o}_n(\xi) \in H^n(B; \pi_{n-1}(V_1(F)))$$

Theorem 3.4.7. *If ξ is an oriented n -plane bundle over a CW complex, then $\mathfrak{o}_n(\xi)$ is equal to the Euler class $e(\xi)$.*

Proof. Using the projection map $\pi_0 : E_0 \rightarrow B$, we can get a pullback bundle $\pi_0^* \xi$ over E_0 . Clearly this pullback bundle has a nowhere zero section, hence

$$\pi_0^* \mathfrak{o}_n(\xi) = \mathfrak{o}_n(\pi_0^* \xi) = 0$$

Using the Gysin exact sequence

$$0 \rightarrow H^0(B) \xrightarrow{\sim e} H^n(B) \xrightarrow{\pi_0^*} H^n(E_0) \rightarrow \cdots$$

with integer coefficients, it follows that

$$\mathfrak{o}_n(\xi) = \lambda \smile e(\xi)$$

for some $\lambda \in H^0(B)$. In particular this argument applies to the universal bundle $\widetilde{\gamma}^n$ over $\widetilde{\text{Gr}}_n$. Using the Gysin sequence

$$0 \rightarrow H^0(\widetilde{\text{Gr}}_n) \xrightarrow{\smile e} H^n(\widetilde{\text{Gr}}_n) \xrightarrow{\pi_0^*} H^n(E_0(\widetilde{\gamma}^n)) \rightarrow \cdots$$

it follows that

$$\mathfrak{o}_n(\widetilde{\gamma}^n) = \lambda_n e(\widetilde{\gamma}^n)$$

for some integer λ_n . Therefore, by naturality,

$$\mathfrak{o}_n(\xi) = \lambda_n e(\xi)$$

for every oriented n -plane bundle ξ over a CW complex.

Now reduce both sides of this equation modulo 2, obtaining

$$w_n(\widetilde{\gamma}^n) = \lambda_n w_n(\widetilde{\gamma}^n)$$

by proposition 3.2.8 and theorem 3.4.1, this proves that λ_n is odd.

If n is odd, then the Euler class itself has order 2 by proposition 3.2.7. Hence $\mathfrak{o}_n(\xi) = e(\xi)$.

If n is even, we must prove that $\lambda_n = 1$. Let τ be the tangent bundle of the n -sphere with n even. Then the Kronecker index $\langle e(\tau), \mu \rangle$ is equal to the Euler characteristic $\chi(S^n) = 2$ by theorem 3.3.13. An analogue formula

$$\langle \mathfrak{o}_n(\xi), \mu \rangle = 2$$

implies that $\lambda_n = 1$. □

3.5. Chern class.

Now we start to talk about Chern classes for complex vector bundles. It is similar to the Stiefel-Whitney classes for real vector bundles.

Let ω be a complex n -plane bundle over B , then ω has a canonical preferred orientation, thus the underlying real vector bundle $\omega_{\mathbb{R}}$ is orientable, then the Euler class

$$e(\omega_{\mathbb{R}}) \in H^{2n}(B; \mathbb{Z})$$

is well-defined.

Now we give an inductive definition of Chern classes for a complex n -plane bundle ω . First we want to construct a canonical $(n-1)$ -plane bundle ω_0 over E_0 . A point in E_0 is specified by a fiber F of ω together with a non-zero vector v in that fiber. Then the fiber of ω_0 over v is by definition, the orthogonal complement of v in the vector space F (We may first put a Hermitian metric on ω). This is a complex vector bundle of dimension $n-1$.

By Gysin sequence 3.4.2, we have a long exact sequence

$$\cdots \rightarrow H^{i-2n}(B) \xrightarrow{\smile e} H^i(B) \xrightarrow{\pi_0^*} H^i(E_0) \rightarrow H^{i-2n+1}(B) \rightarrow \cdots$$

For $i < 2n-1$, $H^{i-2n}(B)$ and $H^{i-2n+1}(B)$ are zero, so it follows that $\pi_0^*: H^i(B) \rightarrow H^i(E_0)$ is an isomorphism.

Definition 3.5.1. *The Chern class $c_i(\omega) \in H^{2i}(B; \mathbb{Z})$ are defined as follows, by induction on the complex dimension n of ω . The **top Chern class** $c_n(\omega)$ is equal to the Euler class $e(\omega_{\mathbb{R}})$. For $i < n$, we set*

$$c_i(\omega) = \pi_0^{*-1} c_i(\omega_0)$$

Finally, for $i \geq n$ the class $c_i(\omega)$ is defined to be zero.

Definition 3.5.2. The formal sum $c(\omega) = 1 + c_1(\omega) + \cdots + c_n(\omega)$ in the ring $H^{\mathbb{N}}(B; \mathbb{Z})$ is called the **total Chern class** of ω .

Lemma 3.5.3 (Naturality). If $f : B \rightarrow B'$ is covered by a bundle map from the complex n -plane bundle ω over B to the complex n -plane bundle ω' over B' , then $c(\omega) = f^*c(\omega')$.

Proof. The Euler class is natural, and it follows from induction. $\square$

Lemma 3.5.4. If ε^k is the trivial complex k -bundle over B , then $c(\omega \oplus \varepsilon^k) = c(\omega)$.

Proof. It suffices to consider the special case $k = 1$, since the genral case follows by induction. Let $\phi = \omega \oplus \varepsilon^1$. Since ϕ has a non-zero section, it follows by proposition 3.2.10 that the top Chern class $c_{n+1}(\phi) = e(\phi_{\mathbb{R}}) = 0$, and hence equal to $c_{n+1}(\omega)$. Let $s : B \rightarrow E_0(\omega \oplus \varepsilon^1)$ be the obvious section. Clearly s is covered by a bundle map $\omega \rightarrow \phi_0$, hence

$$s^*c_i(\phi_0) = c_i(\omega)$$

by naturality. Substituting $\pi_0^*c_i(\phi)$ for $c_i(\phi_0)$, we get the lemma. $\square$

Similar to the real case, we define the **complex Grassmann manifold** $\text{Gr}_n(\mathbb{C}^{n+k})$ to be the set of all complex n -planes through the origin in $\mathbb{C}^{n+k}$. It has a natural structure as smooth manifolds of complex dimension nk . Furthermore there is a canonical complex n -plane bundle which we denote by $\gamma^n = \gamma^n(\mathbb{C}^{n+k})$ over $\text{Gr}_n(\mathbb{C}^{n+k})$. First we study the case $n = 1$.

Theorem 3.5.5. The cohomology ring $H^*(\mathbb{CP}^k; \mathbb{Z})$ is a truncated polynomial ring terminating in dimension $2k$, and generated by the Chern class $c_1(\gamma^1(\mathbb{C}^{k+1}))$.

Proof. Applying the Gysin sequence 3.4.2 to the canonical bundle $\gamma^1 = \gamma^1(\mathbb{C}^{k+1})$ over $\mathbb{CP}^k$, and using the fact that $e_1(\gamma^1) = e(\gamma_{\mathbb{R}}^1)$, we have

$$\cdots \rightarrow H^{i+1}(E_0) \rightarrow H^i(\mathbb{CP}^k) \xrightarrow{\simeq e_1} H^{i+2}(\mathbb{CP}^k) \xrightarrow{\pi_0^*} H^{i+2}(E_0) \rightarrow \cdots$$

Note that E_0 can be identified with $\mathbb{C}^{k+1} \setminus \{0\}$, and hence has the same homotopy type as S^{2k+1} . Thus

$$H^{2k}(\mathbb{CP}^k) \cong \cdots \cong H^2(\mathbb{CP}^k) \cong H^0(\mathbb{CP}^k) \cong \mathbb{Z}$$

generated by $c_1(\gamma^1)^i$ for $i \leq k$. Similarly,

$$H^{2k-1}(\mathbb{CP}^k) \cong \cdots \cong H^{-1}(\mathbb{CP}^k) = 0$$

$\square$

Now let k tend to infinity. The canonical n -plane bundle $\gamma^n(\mathbb{C}^\infty)$ over $\text{Gr}_n(\mathbb{C}^\infty)$ will be denoted by γ^n .

Theorem 3.5.6. The cohomology ring $H^*(\text{Gr}_n(\mathbb{C}^\infty); \mathbb{Z})$ is the polynomial ring over $\mathbb{Z}$ generated by the Chern classes $c_1(\gamma^n), \dots, c_n(\gamma^n)$. There are no polynomial relations between these n generators.

Proof. We prove by induction on n . Since $n = 1$ follows from theorem 3.5.5, we may assume $n \geq 2$. Consider the Gysin sequence 3.4.2

$$\cdots \rightarrow H^i(\text{Gr}_n) \xrightarrow{\simeq c_n} H^{i+2n}(\text{Gr}_n) \xrightarrow{\pi_0^*} H^{i+2n}(E_0) \rightarrow H^{i+1}(\text{Gr}_n) \rightarrow \cdots$$

associated with the bundle γ^n .

We will first show that the cohomology ring $H^*(E_0)$ can be identified with $H^*(\text{Gr}_{n-1})$. In fact a canonical map $f : E_0 \rightarrow \text{Gr}_{n-1}$ is constructed as follows. By definition, a point (X, v) in E_0 consists of an n -plane X in $\mathbb{C}^\infty$ together with a non-zero vector $v \in X$. Let $f(X, v) = X \cap v^\perp$ be the orthogonal complement of v in X . Then $f(X, v)$ is a well-defined $(n-1)$ -plane in $\mathbb{C}^\infty$.

In order to show that f induces cohomology isomorphisms, it is convenient to pass to the sub-bundle $\gamma^n(\mathbb{C}^N) \subseteq \gamma^n$ where N is large. Let $f_N : E_0(\gamma^n(\mathbb{C}^N)) \rightarrow \text{Gr}_{n-1}(\mathbb{C}^N)$ be the corresponding restriction of f . For any $(n-1)$ -plane Y in $\text{Gr}_{n-1}(\mathbb{C}^N)$ it is evident that the inverse image

$$f_N^{-1}(Y) \subseteq E_0(\gamma^n(\mathbb{C}^N))$$

consists of all pairs (X, v) where $v \in \mathbb{C}^N$ is a non-zero vector perpendicular to Y , and where $X = Y + \mathbb{C}v$ is determined by v and Y . Thus f_N can be identified with the projection map

$$E_0(\gamma^{N-n+1}) \rightarrow \text{Gr}_{n-1}(\mathbb{C}^N)$$

where ω^{N-n+1} is the complex vector bundle whose fiber, over $Y \in \text{Gr}_{n-1}(\mathbb{C}^N)$, is the orthogonal complement of Y in $\mathbb{C}^N$.

Using the Gysin sequence 3.4.2 of this new vector bundle, it follows that f_N induces cohomology isomorphisms in dimension $\leq 2(N-n)$. Therefore, taking the direct limit as N tends to infinity, f induces cohomology isomorphisms in all dimensions.

Thus we can insert Gr_{n-1} in place of E_0 in the Gysin sequence, obtaining a new exact sequence of the form

$$\cdots \rightarrow H^i(\text{Gr}_n) \rightarrow H^{i+2n}(\text{Gr}_n) \xrightarrow{\lambda} H^{i+2n}(\text{Gr}_{n-1}) \rightarrow H^{i+1}(\text{Gr}_n) \rightarrow \cdots$$

where $\lambda = f^{*-1}\pi_0^*$.

We must show that this homomorphism $\lambda = f_0^{*-1}\pi_0^*$ maps the Chern class $c_i(\gamma^n)$ to $c_i(\gamma^{n-1})$. This statement is clear for $i = n$, so we may assume that $i \leq n$. By the definition of Chern classes, the image $\pi_0^*c_i(\gamma^n)$ is equal to $c_i(\gamma_0^n)$. But $f : E_0 \rightarrow \text{Gr}_{n-1}$ is clearly covered by a bundle map $\gamma_0^n \rightarrow \gamma^{n-1}$. Therefore $f^*c_i(\gamma^{n-1}) = c_i(\gamma_0^n)$, and it follows that

$$\lambda c_i(\gamma^n) = f^{*-1}\pi_0^*c_i(\gamma^n)$$

is equal to $c_i(\gamma^{n-1})$.

Now let us apply the induction hypothesis. Since $H^*(\text{Gr}_{n-1})$ is generated by the Chern classes $c_1(\gamma^{n-1}), \dots, c_{n-1}(\gamma^{n-1})$, it follows that the homomorphism λ is surjective, so our sequence reduces to

$$0 \rightarrow H^i(\text{Gr}_n) \xrightarrow{\sim c_n} H^{i+2n}(\text{Gr}_n) \xrightarrow{\lambda} H^{i+2n}(\text{Gr}_{n-1}) \rightarrow 0$$

Using this sequence, we will prove, by induction on i , that every element x of $H^{i+2n}(\text{Gr}_n)$ can be expressed **uniquely** as a polynomial in the Chern classes $c_1(\gamma^n), \dots, c_n(\gamma^n)$. Certainly the image $\lambda(x)$ can be expressed uniquely as a polynomial $p(c_1(\gamma^{n-1}), \dots, c_{n-1}(\gamma^{n-1}))$ by our induction hypothesis. Therefore $x - p(c_1(\gamma^n), \dots, c_{n-1}(\gamma^n))$ belongs to the kernel of λ , and hence can be expressed as a product $yc_n(\gamma^n)$ for some uniquely determined $y \in H^i(\text{Gr}_n)$. Now y can be expressed uniquely as a polynomial $q(c_1(\gamma^n), \dots, c_n(\gamma^n))$ by our induction on i , hence

$$x = p(c_1(\gamma^n), \dots, c_{n-1}(\gamma^n)) + c_n(\gamma^n)q(c_1(\gamma^n), \dots, c_n(\gamma^n))$$

The polynomials are unique, since we see that p is unique by applying λ , and q is unique by dividing the difference by $c_n(\gamma^n)$. $\square$

Similar to theorem 3.1.31, we can prove:

Theorem 3.5.7. *Every complex n -bundle over a paracompact base space possesses a bundle map into the canonical complex n -bundle $\gamma^n = \gamma^n(\mathbb{C}^\infty)$ over $\text{Gr}_n = \text{Gr}_n(\mathbb{C}^\infty)$.*

In fact, one can actually prove the sharper statement that **two induced bundles $f^*(\gamma^n)$ and $g^*(\gamma^n)$ are isomorphic if and only if f is homotopic to g .** For this reason, $\text{Gr}_n(\mathbb{C}^\infty)$ is called the **classifying space** for complex n -plane bundles, and denoted by $BU(n)$.

And just as the real case, we have the product theorem for Chern classes, with more effort.

Theorem 3.5.8. *ω and ϕ are two complex bundles over the same base space, then*

$$c(\omega \oplus \phi) = c(\omega)c(\phi)$$

If ω is a complex vector bundle, the **conjugate bundle** $\bar{\omega}$ is defined to be the complex vector bundle with the same underlying real vector bundle

$$\omega_{\mathbb{R}} = \bar{\omega}_{\mathbb{R}}$$

but with the opposite complex structure. Thus, the identity map $f : E(\omega) \rightarrow E(\bar{\omega})$ is conjugate linear,

$$f(\lambda e) = \bar{\lambda} f(e)$$

for every complex number λ and every $e \in E(\omega)$. In particular, it follows that $f(ie) = -if(e)$.

Lemma 3.5.9. *The Chern class $c_k(\bar{\omega})$ is equal to $(-1)^k c_k(\omega)$. Hence*

$$c(\bar{\omega}) = 1 - c_1(\omega) + c_2(\omega) - \cdots \pm c_n(\omega)$$

Proof. Note that the two oriented real vector bundle $\omega_{\mathbb{R}}$ and $\bar{\omega}_{\mathbb{R}}$ have the same orientation if n is even, but the opposite orientation if n is odd. It follows that the top Chern class

$$c_n(\omega) = e(\omega_{\mathbb{R}})$$

is equal to $(-1)^n c_n(\bar{\omega})$. Then it follows from induction. $\square$

Closely related to the conjugate bundle $\bar{\omega}$ is the **dual bundle** $\text{Hom}_{\mathbb{C}}(\omega, \mathbb{C})$. By definition this is the complex vector bundle over the same base space whose typical fiber is equal to the dual $\text{Hom}_{\mathbb{C}}(F, \mathbb{C})$.

If ω possesses a Hermitian metric, then $\text{Hom}(\omega, \mathbb{C})$ is canonically isomorphic to the conjugate bundle $\bar{\omega}$. This is because if we are given a Hermitian inner product

$$\langle v_1, v_2 \rangle \in \mathbb{C}$$

on the typical fiber F , linear in the first variable and conjugate linear in the second, then the correspondence

$$v_2 \mapsto \langle -, v_2 \rangle$$

maps the conjugate vector space $\bar{F}$ isomorphically to the dual vector space $\text{Hom}(F, \mathbb{C})$.

As an application, consider the tangent bundle τ^n of $\mathbb{CP}^n$.

Theorem 3.5.10. *The total Chern class $c(\tau^n)$ is equal to $(1 + a)^{n+1}$ where a is a suitable chosen generator for $H^2(\mathbb{CP}^n; \mathbb{Z})$.*

Proof. Let $\gamma^1 = \gamma^1(\mathbb{CP}^{n+1})$ be the canonical line bundle over $\mathbb{CP}^n$, and let ω^n be its orthogonal complement, so that the Whitney sum $\gamma^1 \oplus \omega^n$ is a trivial complex $(n+1)$ -plane bundle over $\mathbb{CP}^n$. We will show that the complex vector bundle

$$\text{Hom}_{\mathbb{C}}(\gamma^1, \omega^n)$$

can be identified with the tangent bundle τ^n of $\mathbb{CP}^n$. In fact if L is a complex line through the origin in $\mathbb{C}^{n+1}$, and $L^\perp$ is its orthogonal complement, then $\text{Hom}(L, L^\perp)$ can be identified with the neighborhood of L in $\mathbb{CP}^n$ consisting of all lines L' which can be considered as graphs of linear maps from L to $L^\perp$. It follows that the tangent space of $\mathbb{CP}^n$ at L can be identified with $\text{Hom}(L, L^\perp)$, and hence that $\tau^n \cong \text{Hom}(\gamma^1, \omega^n)$.

Now adding the trivial line bundle $\varepsilon^1 \cong \text{Hom}(\gamma^1, \gamma^1)$ to both sides, it follows that

$$\tau^n \oplus \varepsilon^1 \cong \text{Hom}(\gamma^1, \omega^n \oplus \gamma^1) \cong \text{Hom}(\gamma^1, \varepsilon^1 \oplus \cdots \oplus \varepsilon^1)$$

Clearly this can be identified with the Whitney sum of $n+1$ copies of the dual bundle $\text{Hom}(\gamma^1, \varepsilon^1) \cong \bar{\gamma}^1$. Thus the total Chern class $c(\tau^n \oplus \varepsilon^1)$ is equal to

$$c(\bar{\gamma}^1)^{n+1} = (1 - c_1(\gamma^1))^{n+1}$$

Setting $a = -c_1(\gamma^1)$, the conclusion follows. $\square$

Now we can give an equivalent definition of Chern classes using axiom system:

Theorem 3.5.11. *We give the following axiom system for Chern classes:*

Axiom 1 *To each complex vector bundle ω there corresponds a sequence of cohomology class*

$$c_i(\omega) \in H^{2i}(B; \mathbb{Z}), \quad i = 0, 1, 2, \dots$$

*called the **Chern classes** of ω . The class $c_0(\omega)$ is equal to the unit element*

$$1 \in H^0(B; \mathbb{Z})$$

and $c_i(\omega)$ equals zero for $i > n$ if ω is an n -plane bundle.

Axiom 2 (Naturality) *If $f : B(\omega) \rightarrow B(\eta)$ is covered by a bundle map from ω to η , then*

$$c_i(\omega) = f^*(c_i(\eta))$$

Axiom 3 (Whitney product theorem) *If ω and η are vector bundles over the same base space, then*

$$c_k(\omega \oplus \eta) = \sum_{i=0}^k c_i(\omega) \smile c_{k-i}(\eta)$$

Axiom 4 *For the line bundle ω over the $\mathbb{CP}^1$, the Chern class $c_1(\omega)$ is non-zero.*

Proof. We have proved the existence and uniqueness is the same as theorem 3.1.41. $\square$

3.6. Pontrjagin class.

To obtain further information about real vector bundles, we consider the following construction. For a real n -plane bundle ξ , we obtain a complex n -plane bundle with typical fiber $F \otimes \mathbb{C}$. We denote this new bundle by $\xi \otimes \mathbb{C}$ and call it the **complexification** of ξ .

Lemma 3.6.1. *The complexification $\xi \otimes \mathbb{C}$ of ξ is always isomorphic to its own conjugate bundle $\overline{\xi \otimes \mathbb{C}}$.*

Proof. $f : x + iy \mapsto x - iy$ gives the isomorphism. $\square$

Now consider the total Chern class

$$c(\xi \otimes \mathbb{C}) = 1 + c_1(\xi \otimes \mathbb{C}) + c_2(\xi \otimes \mathbb{C}) + \cdots + c_n(\xi \otimes \mathbb{C})$$

of this complexified n -plane bundle. Setting this equal to

$$c(\overline{\xi \otimes \mathbb{C}}) = 1 - c_1(\xi \otimes \mathbb{C}) + c_2(\xi \otimes \mathbb{C}) - \cdots \pm c_n(\xi \otimes \mathbb{C})$$

we see that the odd Chern classes

$$c_1(\xi \otimes \mathbb{C}), c_3(\xi \otimes \mathbb{C}), \dots$$

are all elements of order 2.

Definition 3.6.2. *Ignoring these elements of order 2, the i -th **Pontrjagin class***

$$p_i(\xi) \in H^{4i}(B; \mathbb{Z})$$

*is defined to be the integral cohomology class $(-1)^i c_{2i}(\xi \otimes \mathbb{C})$. Evidently $p_i(\xi)$ is zero for $i > n/2$. The **total Pontrjagin class** is defined to be unit*

$$p(\xi) = 1 + p_1(\xi) + \cdots + p_{[n/2]}(\xi)$$

in the ring $H^{\Pi}(B; \mathbb{Z})$.

Lemma 3.6.3. *Pontrjagin classes are natural with respect to bundle maps. Furthermore, if ε^k is a trivial k -plane bundle, then $p(\xi \oplus \varepsilon^k) = p(\xi)$.*

Proof. This follows from lemma 3.5.3 and 3.5.4. $\square$

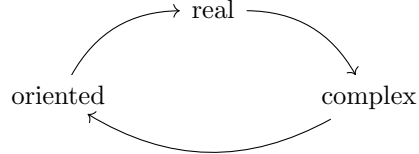
The product theorem for Pontrjagin classes is as follows (since we have thrown away odd terms, we can only get a weaker statement.)

Theorem 3.6.4. *$p(\xi \oplus \eta)$ is congruent to $p(\xi)p(\eta)$ modulo elements of order 2. In other words*

$$2(p(\xi \oplus \eta) - p(\xi)p(\eta)) = 0$$

Proof. This follows from the product theorem of Chern classes. $\square$

Real bundles, complex bundles, and oriented bundles can be transformed as follows:



If we start with a real n -plane bundle ξ , we can first form the induced complex n -plane bundle $\xi \otimes \mathbb{C}$. Then, forgetting the complex structure, we obtain the underlying real $2n$ -plane bundle $(\xi \otimes \mathbb{C})_{\mathbb{R}}$ with a canonical preferred orientation. Finally,

forgetting the orientation, this resulting real $2n$ -plane is isomorphic to $\xi \oplus \xi$.

Now if we start with a complex vector bundle ω , we have the following:

Lemma 3.6.5. *For any complex vector bundle ω , the complexification $\omega_{\mathbb{R}} \otimes \mathbb{C}$ of the underlying real vector bundle is canonically isomorphic to $\omega \oplus \bar{\omega}$.*

Proof. Suppose that $V = F_{\mathbb{R}}$ where F is the typical fiber of ω . Then the correspondence

$$g : x \mapsto (x, -ix)$$

from F to $V \oplus V$ is complex linear, that $g(ix) = J(g(x))$. Similarly the correspondence

$$h : x \mapsto (x, ix)$$

from F to $V \oplus V$ is conjugate linear. Since every point (x, y) of $V \oplus V \cong F_{\mathbb{R}} \otimes \mathbb{C}$ can be written uniquely as the sum

$$g\left(\frac{x+iy}{2}\right) + h\left(\frac{x-iy}{2}\right)$$

it follows that $F_{\mathbb{R}} \otimes \mathbb{C}$ is canonically isomorphic, as complex vector space, to $F \oplus \bar{F}$. Hence $\omega_{\mathbb{R}} \otimes \mathbb{C} \cong \omega \oplus \bar{\omega}$. $\square$

Corollary 3.6.6. *For any complex n -plane bundle ω , the Chern classes $c_i(\omega)$ determine the Pontrjagin classes $p_k(\omega_{\mathbb{R}})$ by the formula*

$$1 - p_1 + p_2 - \cdots \pm p_n = (1 - c_1 + c_2 - \cdots \pm c_n)(1 + c_1 + c_2 + \cdots + c_n)$$

Proof. This follows from theorem 3.5.8 and lemma 3.5.9. $\square$

Example 3.6.7. *Let τ be the tangent bundle of $\mathbb{CP}^n$. Since the total Chern class $c(\tau)$ equals $(1+a)^{n+1}$ by theorem 3.5.10, it follows that the Pontrjagin classes $p_k(\tau_{\mathbb{R}})$ are given by*

$$1 - p_1 + \cdots \pm p_n = (1-a)^{n+1}(1+a)^{n+1} = (1-a^2)^{n+1}$$

Therefore the total Pontrjagin class $1 + p_1 + \cdots + p_n$ is equal to $(1+a^2)^{n+1}$. In other words

$$p_k(\mathbb{CP}^n) = \binom{n+1}{k} a^{2k}$$

for $1 \leq k \leq n/2$. Here we are using $p_k(M)$ for the Pontrjagin class $p_k(\tau(M)_{\mathbb{R}})$ of a complex manifold M .

Now suppose we start with an oriented n -plane bundle ξ , we have the following

Lemma 3.6.8. *The real $2n$ -plane bundle $(\xi \otimes \mathbb{C})_{\mathbb{R}}$ is isomorphic to $\xi \oplus \xi$ under an isomorphism which either preserves or reverses orientation according as $n(n-1)/2$ is even or odd.*

Proof. Suppose $v_1, \dots, v_n$ is an ordered basis for ξ , then the orientation for $(\xi \otimes \mathbb{C})_{\mathbb{R}}$ is $v_1, iv_1, \dots, v_n, iv_n$, but the orientation for $\xi \oplus \xi$ is $v_1, \dots, v_n, v_1, \dots, v_n$. $\square$

Corollary 3.6.9. *If ξ is an oriented $2k$ -plane bundle, then the Pontrjagin class $p_k(\xi)$ is equal to the square of the Euler class $e(\xi)$.*

Proof. This follows from lemma 3.6.8. $\square$

4. CHERN-WEIL THEORY

Now we turn the second part of this paper. We will turn to geometric construction of Chern classes, which is called the Chern-Weil theory. The main tool we use is connection on vector bundles. We refer to [25] in this section.

4.1. Connection and curvature on vector bundles.

In the following, we consider smooth vector bundles with coefficients $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.

Before talking about connection, we quickly mention the concept of vector-valued differential forms. Given a vector bundle ξ over M , we define the **vector bundle of ξ -valued differential s -forms** as the tensor product $\bigwedge^s \tau^* \otimes \xi$, where τ is the tangent bundle of M , and the spaces of sections are denoted by

$$\Omega^s(U, \xi) = \Gamma^\infty(U, \bigwedge^s \tau^* \otimes \xi) = \Omega^s(U) \otimes_{C^\infty(U)} \Gamma^\infty(U, \xi)$$

If ξ is a trivial bundle, with fiber some fixed vector space V , then we also write $\Omega^s(\cdot, V)$ for the ξ -valued forms, which we call **V -valued forms**.

Given a vector space V with a bilinear map $\mu : V \times V \rightarrow V$, then we can define a wedge product for V -valued forms by

$$(\omega \wedge_\mu \tau)(v_1, \dots, v_{s+r}) = \frac{1}{s!r!} \sum_{\sigma \in \mathfrak{S}_{s+r}} \text{sgn}(\sigma) \mu(\omega(v_{\sigma(1)}, \dots, v_{\sigma(s)}), \tau(v_{\sigma(s+1)}, \dots, v_{\sigma(s+r)}))$$

This will be of importance in the following cases: If V is some space of square matrices and μ is the usual matrix multiplication, we simply omit the subscript μ , and for multiplication with 0-forms, we even omit the wedge symbol. If V is some Lie algebra and $\mu = [\cdot, \cdot]$ is its Lie bracket, then instead we write $[\omega \wedge \tau]$.

The notion of the exterior derivative also extends to vector-valued forms, by simply defining

$$d(\omega \otimes s) = d(\omega) \otimes s$$

where $\omega \in \Omega^s(U)$ and $s \in \Gamma^\infty(U, \xi)$. Thus, we get a map

$$d : \Omega^s(U, \xi) \rightarrow \Omega^{s+1}(U, \xi)$$

Definition 4.1.1. Let ξ be a smooth vector bundle over M . A **connection** on ξ is given by a $\mathbb{K}$ -bilinear map

$$\nabla : \mathfrak{X}(M) \times \Gamma^\infty(M, \xi) \rightarrow \Gamma^\infty(M, \xi), (X, s) \mapsto \nabla_X s$$

such that for any $f \in C^\infty(M)$ we have

- (1) $\nabla_{fX} s = f \nabla_X s$.
- (2) $\nabla_X (fs) = X(f)s + f \nabla_X s$.

We will also refer to the first property as **tensoriality** of ∇_X in the X . Tensoriality implies that one can alternatively think of a connection as a $\mathbb{K}$ -linear map

$$\nabla : \Omega^0(M, \xi) \rightarrow \Omega^1(M, \xi)$$

satisfying the Leibniz rule

$$\nabla(fs) = df \otimes s + f \nabla(s)$$

Lemma 4.1.2. ∇ is local in the following sense: $\nabla(s)_x$ only depends the value of s in a small neighborhood of x .

Proof. Suppose $s_1, s_2 \in \Omega^0(M, \xi)$ agree in a neighborhood U of $x \in M$. Pick a bump function $\lambda \in C^\infty(M)$ that is constantly 1 in a smaller neighborhood of x and whose support lies in U . Then $\lambda(s_1 - s_2) = 0$, and

$$\begin{aligned} 0 &= \nabla(\lambda(s_1 - s_2))_x = \nabla(\lambda s_1)_x - \nabla(\lambda s_2)_x \\ &= \lambda(x)(\nabla(s_1)_x - \nabla(s_2)_x) = \nabla(s_1)_x - \nabla(s_2)_x \end{aligned}$$

since $d\lambda(x) = 0$. $\square$

Proposition 4.1.3. *Any smooth vector bundle ξ over a smooth manifold M admits a connection.*

Proof. Given a local frame $e_1, \dots, e_n$ on some $U \subseteq M$ we can define a connection ∇_U on $\xi|_U$ by defining the e_i to be flat, and extending by the Leibniz rule to

$$\nabla_U(s^i e_i) = ds^i \otimes e_i$$

If one covers M by open subsets $\{U_\alpha\}$, then one can glue the connections ∇_U using a partition of unity subordinate to $\{U_\alpha\}$. $\square$

Definition 4.1.4. *Given a connection ∇ on ξ , we define its **curvature tensor** R as the multilinear map $\mathfrak{X}(M) \times \mathfrak{X}(M) \times \Gamma^\infty(M, \xi) \rightarrow \Gamma^\infty(M, \xi)$*

$$R(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s$$

We can easily check that R is tensorial in each component, and alternating in X and Y , so this defines a $C^\infty(M)$ -linear map

$$R : \Omega^0(M, \xi) \rightarrow \Omega^2(M, \xi)$$

or equivalently, a vector-valued 2-form

$$R \in \Omega^2(M, \text{Hom}(\xi, \xi))$$

Now we try to express connection and curvature in local coordinates. Given a local frame $e = (e_1, \dots, e_n) \in \Omega^0(I, \xi)^n$, we can express the connection as

$$\nabla e_j = \omega_j^i e_i$$

where $(\omega_j^i)_{i,j}$ is an $(n \times n)$ -matrix of $\mathbb{K}$ -valued 1-forms. Equivalently, we have a $\mathfrak{gl}(n, \mathbb{K})$ -valued differential form $\omega \in \Omega^1(U, \mathfrak{gl}(n, \mathbb{K}))$ with components ω_j^i . We call ω the **connection form relative to the frame** e . In a similar way, we can express the curvature form as

$$R e_j = \Omega_j^i e_i$$

where $\Omega \in \Omega^2(U, \mathfrak{gl}(n, \mathbb{K}))$ with entries Ω_j^i , which we also call the **curvature form relative to** e .

Definition 4.1.5. *Given a Riemannian or Hermitian metric $\langle \cdot, \cdot \rangle$ on ξ , we call a connection ∇ on ξ **compatible with the metric** if*

$$d\langle s_1, s_2 \rangle = \langle \nabla s_1, s_2 \rangle + \langle s_1, \nabla s_2 \rangle \in \Omega^1(M, \mathbb{K})$$

for all (local) sections s_1 and s_2 .

Similar to proposition 4.1.3, we can prove that

Proposition 4.1.6. *Any smooth Riemannian or Hermitian vector bundle admits a compatible connection.*

Now suppose e is a local orthonormal frame with respect to the given metric. Then the compatibility gives

$$\begin{aligned} 0 &= \langle \nabla e_i, e_j \rangle + \langle e_i, \nabla e_j \rangle \\ &= \langle e_k, e_j \rangle \omega_i^k + \langle e_i, e_k \rangle \omega_j^k \\ &= \omega_j^i + \omega_i^j \end{aligned}$$

so the connection matrix ω_j^i with respect to the orthonormal frame is skew-symmetric.

The followings are some properties of matrix-valued differential forms.

Proposition 4.1.7. *Let $A \in \Omega^s(U, \mathfrak{gl}(n, \mathbb{K}))$, $B \in \Omega^r(U, \mathfrak{gl}(n, \mathbb{K}))$. Denote by A^T the usual transposition, by $A \wedge B$ the wedge product associated to matrix multiplication, and by $\text{tr}(A)$ the $\mathbb{K}$ -valued s -form that is the sum of diagonal entries of A .*

- (1) $(A \wedge B)^T = (-1)^{sr} B^T \wedge A^T$.
- (2) $\text{tr}(A \wedge B) = (-1)^{sr} \text{tr}(B \wedge A)$.
- (3) $d\text{tr}(A) = \text{tr}(dA)$.
- (4) $d(A \wedge B) = (dA) \wedge B + (-1)^s A \wedge (dB)$.

Theorem 4.1.8 (Cartan's structure equation). *Suppose ω is the connection form and Ω is the curvature form relative to some local frame on $U \subseteq M$, then*

$$\Omega = d\omega + \omega \wedge \omega$$

Proof. We use the Einstein summation convention.

$$\begin{aligned} R(X, Y)e_j &= \nabla_X \nabla_Y e_j - \nabla_Y \nabla_X e_j - \nabla_{[X, Y]} e_j \\ &= \nabla_X (\omega_j^l(Y) e_l) - \nabla_Y (\omega_j^l(X) e_l) - \omega_j^l([X, Y]) e_l \\ &= X(\omega_j^l(Y)) e_l + \omega_j^l(Y) \nabla_X e_l \\ &\quad - Y(\omega_j^l(X)) e_l - \omega_j^l(X) \nabla_Y e_l - \omega_j^l([X, Y]) e_l \\ &= d\omega_j^l(X, Y) e_l + \omega_j^l(Y) \omega_l^k(X) e_k - \omega_j^l(X) \omega_l^k(Y) e_k \\ &= (d\omega + \omega \wedge \omega)_j^l(X, Y) e_l \end{aligned}$$

Hence $\Omega = d\omega + \omega \wedge \omega$. □

Corollary 4.1.9 (second Bianchi identity). *For any $k \geq 1$, we have*

$$d(\Omega^k) = \Omega^k \wedge \omega - \omega \wedge \Omega^k$$

where $\Omega^k = \Omega \wedge \cdots \wedge \Omega$ is the k -fold wedge product of Ω with itself.

Proof. Induction on k . □

Now let's examine how the relative expression for connection and curvature forms transform under change of frame. Suppose we have a second frame $\bar{e}_j = \sum_i a_j^i e_i$, or short, $\bar{e} = ea$, $a \in C^\infty(U, \text{GL}(n, \mathbb{K}))$. Then, using the Leibniz rule

$$\begin{aligned} \nabla_X \bar{e}_j &= \nabla_X (a_j^l e_l) \\ &= X(a_j^l) e_l + a_j^l \nabla_X (e_l) \\ &= da_j^l(X) (a^{-1})_l^i \bar{e}_i + a_j^l \omega_l^k(X) (a^{-1})_k^i \bar{e}_i \end{aligned}$$

thus

$$\bar{\omega}_j^i = (a^{-1})_l^i da_j^l + (a^{-1})_k^i \omega_l^k a_j^l$$

or briefly,

$$\bar{\omega} = a^{-1}da + a^{-1}\omega a$$

We can do the same thing for the curvature form

$$\begin{aligned} R(X, Y)\bar{e}_j &= R(X, Y)(a_j^l e_l) \\ &= a_j^l R(X, Y)e_l \\ &= a_j^l \Omega_l^k(X, Y)e_k \\ &= a_j^l \Omega_l^k(X, Y)(a^{-1})_k^i \bar{e}_i \end{aligned}$$

so

$$\bar{\Omega}_j^i = (a^{-1})_k^i \Omega_l^k a_j^l$$

or briefly,

$$\bar{\Omega} = a^{-1}\Omega a$$

This transformation behavior of the curvature is the starting point of Chern-Weil theory, in particular, the construction of the Chern-Weil homomorphism, which leads to the geometric definition of Chern classes.

4.2. Chern-Weil homomorphism.

Recall that the Lie group $\mathrm{GL}(n, \mathbb{K}) = G$ acts on its Lie algebra $\mathfrak{gl}(n, \mathbb{K}) = \mathfrak{g}$ through the **adjoint action**

$$\mathrm{Ad}_G : G \rightarrow \mathrm{GL}(\mathfrak{g}), \quad \mathrm{Ad}_G(a)(v) = ava^{-1}$$

The previous calculation tells us that *any construction as a function of the curvature matrix Ω that is invariant under the adjoint action is independent of the choice of local frame.*

Now we consider a polynomial P in n^2 variables,

$$P \in \mathbb{K}[\mathfrak{g}]$$

such that $P(\mathrm{Ad}_G(a)(v)) = P(v)$ for all $a \in G$ and $v \in \mathfrak{g}$. We denote the space of such Ad_G -invariant polynomials by

$$\mathbb{K}[\mathfrak{g}]^G$$

A polynomial can be evaluated on the $\mathfrak{g}$ -valued 2-form Ω : Note that the space of even dimension differential forms, denoted by $\Omega^{2*}(U)$, with the wedge product is a commutative algebra, and hence there is an evaluation homomorphism of algebras

$$\mathbb{K}[\mathfrak{g}] \rightarrow \Omega^{2*}(U), \quad P \mapsto P(\Omega)$$

where Ω is defined relative to some local frame on U . If P is Ad_G -invariant, then $P(\Omega)$ does not depend on the local frame, and hence is globally defined. This gives us a homomorphism of algebras

$$C_\nabla : \mathbb{K}[\mathfrak{g}]^G \rightarrow \Omega^{2*}(M) \subseteq \Omega^*(M)$$

which depends only on the connection ∇ .

Regarding the above construction, we have the following two fundamental results.

Theorem 4.2.1. *Let ξ be a smooth $\mathbb{K}$ -vector bundle over a smooth manifold M , ∇ a connection on ξ , and P an $\mathrm{Ad}_{\mathrm{GL}(n, \mathbb{K})}$ -invariant homogeneous polynomial of degree k on $\mathfrak{gl}(n, \mathbb{K})$. Then*

- (1) *the form $C_\nabla(P) \in \Omega^{2k}(M, \mathbb{K})$ is closed, and*
- (2) *the cohomology class of $C_\nabla(P)$ in $H^{2k}(M, \mathbb{K})$ is independent of ∇ .*

We will prove this later. By this theorem, and by existence of connection 4.1.3, we can associate to any vector bundle ξ a natural homomorphism of graded (up to multiplying degree by 2) algebras

$$\begin{aligned} c_\xi : \mathbb{K}[\mathfrak{g}]^G &\longrightarrow H^*(M, \mathbb{K}) \\ P &\longmapsto [C(P)] \end{aligned}$$

This homomorphism is called the **Chern-Weil homomorphism**.

Example 4.2.2. Consider the polynomial

$$P(\Omega) = \det(I + \Omega)$$

then

$$P(\text{Ad}(a)(\Omega)) = \det(I + a\Omega a^{-1}) = \det(a(I + \Omega)a^{-1}) = \det(I + \Omega) = P(\Omega)$$

so $P \in \mathbb{K}[\mathfrak{g}]^G$. Note that P is inhomogeneous of degree n , and denote its homogeneous components by $f_k \in \mathbb{K}[\mathfrak{g}]_k^G$. Then, for example, $f_0 = 1$, and $f_n(\Omega) = \det(\Omega)$.

Example 4.2.3. Another example of non-trivial invariant polynomials is given by the trace polynomials

$$\Sigma_k(\Omega) = \text{tr}(\Omega^k), \quad k = 1 \cdots, n$$

As before, using elementary linear algebra, it can easily be checked that these are Ad_G -invariant.

Theorem 4.2.4. The polynomials f_k , as well as the polynomials Σ_k , are algebraically independent over $\mathbb{K}$ and generate $\mathbb{K}[\mathfrak{g}]^G$ as a $\mathbb{K}$ -algebra:

$$\mathbb{K}[\mathfrak{g}]^G = \mathbb{K}[f_1, \dots, f_n] = \mathbb{K}[\Sigma_1, \dots, \Sigma_n]$$

Proof. Consider an evaluation homomorphism

$$\begin{aligned} \mathbb{K}[\mathfrak{g}] &\longrightarrow \mathbb{K}[t_1, \dots, t_n] \\ P &\longmapsto P(\text{diag}(t_1, \dots, t_n)) \end{aligned}$$

Now if P is invariant, then its value on $\text{diag}(t_1, \dots, t_n)$ must stay the same if we reorder the variables $t_1, \dots, t_n$, as such a reordering can be written as conjugation by a permutation matrix. In other words, the above homomorphism restricts to a homomorphism

$$\Phi : \mathbb{K}[\mathfrak{g}]^G \rightarrow \mathbb{K}[t_1, \dots, t_n]^{\mathfrak{S}_n}$$

to the algebra of symmetric polynomials in n variables over $\mathbb{K}$.

We claim that this homomorphism is an isomorphism. First, injectivity of Φ can be shown as follows: If $\Phi(P) = 0$, then P takes 0 on all diagonal matrices, hence takes 0 on all diagonalizable matrices. But the set of diagonalizable matrices contains an open subset of $\mathfrak{g}$. Thus $P = 0$ as a polynomial on affine space.

Under this inclusion, f_k are mapped to the elementary symmetric polynomials s_k , and the trace polynomials Σ_k correspond to the power sum polynomials $p_k(t_1, \dots, t_n) = \sum_i (t_i)^k$. It is well-known that both s_k and p_k generate $\mathbb{K}[t_1, \dots, t_n]^{\mathfrak{S}_n}$ freely as a $\mathbb{K}$ -algebra. This firstly proves surjectivity of Φ , and secondly proves the theorem. $\square$

Proof. (of theorem 4.2.1 (1)) By proposition 4.1.7 and second Bianchi identity 4.1.9, we have

$$\begin{aligned}
 d(\operatorname{tr}(\Omega^k)) &= \operatorname{tr}(d(\omega^k)) \\
 &= \operatorname{tr}(\Omega^k \wedge \omega - \omega \wedge \Omega^k) \\
 &= \operatorname{tr}(\Omega^k \wedge \omega) - (-1)^{2k}(\Omega^k \wedge \omega) \\
 &= 0
 \end{aligned}$$

This shows that the forms $C_\nabla(\Sigma_k)$ are closed.

By theorem 4.2.4, any other homogeneous form in the image of C can be written as some polynomial in the forms $C_\nabla(\Sigma_k)$, and since the wedge product and linear combinations of closed forms are closed, any such form is closed. $\square$

It remains to prove independence of the connection. First note that any two connections ∇^0 and ∇^1 of ξ can be connected by a family

$$\nabla^t = t\nabla^1 + (1-t)\nabla^0, \quad 0 \leq t \leq 1$$

of connections. Let $P \in \mathbb{K}[\mathfrak{g}]_k^G$. In order to prove that $[C_{\nabla^0}(P)] = [C_{\nabla^1}(P)] \in H^{2k}(M, \mathbb{K})$, it suffices to prove that the cohomology class of $C_{\nabla^t}(P)$ does not vary along t , or equivalently, that

$$\frac{d}{dt} C_{\nabla^t}(P)$$

is an exact differential form for all t .

To be more precise, with respect to some local frame, denote by ω_t the connection form, and by Ω_t the curvature form of ∇^t . Then ω_t and Ω_t are families of differential forms smoothly varying in t .

Let $\{\omega_t\}_{t \in J}$, $J \subseteq \mathbb{R}$ some interval, be any smoothly varying family of $\mathfrak{g}$ -valued k -forms. Then we can differential ω_t by t to get a smoothly varying k -form $\frac{d}{dt}\omega_t = \dot{\omega}_t$. In the same way one can integrate smooth families of differential forms on some interval $[a, b] \subseteq \mathbb{R}$, to get $\int_a^b \omega_t dt$. Here are some basic properties of differentiation and integration of smoothly varying $\mathfrak{g}$ -valued forms.

Proposition 4.2.5. *Let $J \subseteq \mathbb{R}$ be some connected subset, and $\{\omega_t\}_{t \in J}$ and $\{\tau_t\}_{t \in J}$ two smoothly varying $\mathfrak{g}$ -valued forms. Then*

- (1) $\frac{d}{dt} \operatorname{tr}(\omega_t) = \operatorname{tr}(\dot{\omega}_t)$.
- (2) $\frac{d}{dt}(\omega_t \wedge \tau_t) = \dot{\omega}_t \wedge \tau_t + \omega_t \wedge \dot{\tau}_t$.
- (3) $\frac{d}{dt}(d\omega_t) = d(\dot{\omega}_t)$.
- (4) for $[a, b] \subseteq J$, $\int_a^b d\omega_t dt = d \int_a^b \omega_t dt$.

Now take a family of connection ∇^t . With respect to some fixed local frame, denote by ω_t the connection forms and by Ω_t the curvature forms. If $\nabla^t = t\nabla^1 + (1-t)\nabla^0$, then $\omega_t = t\omega_1 + (1-t)\omega_0$, and hence ω_t and $\Omega_t = d\omega_t + \omega_t \wedge \omega_t$ are smoothly varying families of forms.

As before, in order to prove that C_∇ is independent of ∇ , it suffices to prove that $C_\nabla(\Sigma_k)$ are independent of ∇ for $k = 1, \dots, n$. We need a technical result as follows.

Proposition 4.2.6. *If $\{\nabla^t\}_{t \in J}$, $J \subseteq \mathbb{R}$ connected, is a family of connections such that ω_t and Ω_t , with respect to some fixed local frame, depend smoothly on t . Then*

for each $k \geq 1$,

$$\frac{d}{dt} \text{tr}(\Omega_t^k) = d(k \text{tr}(\Omega_t^{k-1} \wedge \dot{\omega}_t))$$

In particular, $\frac{d}{dt} C_{\nabla^t}(\Sigma_k)$ is an exact $\mathbb{K}$ -valued differential form for each k .

Proof. The left hand side can be rewritten as

$$\begin{aligned} \frac{d}{dt} \text{tr}(\Omega_t^k) &= \text{tr}\left(\frac{d}{dt} \Omega_t^k\right) \\ &= \text{tr}(\dot{\Omega}_t \wedge \Omega_t^{k-1} + \Omega_t \wedge \dot{\Omega}_t \wedge \Omega_t^{k-2} + \cdots + \Omega_t^{k-1} \wedge \dot{\Omega}_t) \\ &= \text{tr}(k \Omega_t^{k-1} \wedge \dot{\Omega}_t) \end{aligned}$$

by Leibniz rule. By Cartan's structure equation 4.1.8, we have

$$\dot{\Omega}_t = \frac{d}{dt}(d\omega_t) + \frac{d}{dt}(\omega_t \wedge \omega_t) = d\dot{\omega}_t + \dot{\omega}_t \wedge \omega_t + \omega_t \wedge \dot{\omega}_t$$

hence

$$\begin{aligned} \frac{d}{dt} \text{tr}(\Omega_t^k) &= k \text{tr}(\Omega_t^{k-1} \wedge d\dot{\omega}_t + \Omega_t^{k-1} \wedge \dot{\omega}_t \wedge \omega_t + \Omega_t^{k-1} \wedge \omega_t \wedge \dot{\omega}_t) \\ &= k \text{tr}(\Omega_t^{k-1} \wedge d\dot{\omega}_t + (\Omega_t^{k-1} \wedge \omega_t - \omega_t \wedge \Omega_t^{k-1}) \wedge \dot{\omega}_t) \\ &= k \text{tr}(\Omega_t^{k-1} \wedge d\dot{\omega}_t + d(\Omega_t^{k-1}) \wedge \dot{\omega}_t) \\ &= k \text{tr}(d(\Omega_t^{k-1} \wedge \dot{\omega}_t)) \\ &= d(k \text{tr}(\Omega_t^{k-1} \wedge \dot{\omega}_t)) \end{aligned}$$

where we used the second Bianchi identity 4.1.9. $\square$

Proof. (of theorem 4.2.1(2)) As mentioned before, let ∇^t , $t \in [0, 1]$ be a family of connections such that ω_t and Ω_t are smoothly varying in t .

We want to show the right hand side in proposition 4.2.6 is actually globally defined. For this, suppose $\bar{e} = ea$, $a \in C^\infty(U, \text{GL}(n, \mathbb{K}))$ is another local frame. Then $\bar{\omega}_t = a^{-1}\omega_t a + a^{-1}da$ as shown before, hence

$$\dot{\bar{\omega}}_t = a^{-1}\dot{\omega}_t a$$

Thus

$$\text{tr}(\bar{\Omega}_t^{k-1} \wedge \dot{\bar{\omega}}_t) = \text{tr}(a^{-1} \Omega_t^{k-1} a a^{-1} \dot{\omega}_t a) = \text{tr}(a^{-1} \Omega_t^{k-1} \dot{\omega}_t a) = \text{tr}(\Omega_t^{k-1} \wedge \dot{\omega}_t)$$

Thus by proposition 4.2.6, we have

$$\begin{aligned} \text{tr}(\Omega_1^k) - \text{tr}(\Omega_0^k) &= \int_0^1 \frac{d}{dt} \text{tr}(\Omega_t^k) dt \\ &= \int_0^1 d(k \text{tr}(\Omega_t^{k-1} \wedge \dot{\omega}_t)) dt \\ &= d \int_0^1 k \text{tr}(\Omega_t^{k-1} \wedge \dot{\omega}_t) dt \end{aligned}$$

a globally defined, exact differential form. This means that $C_{\nabla^1}(\Sigma_k) - C_{\nabla^0}(\Sigma_k)$ is exact, so the two homomorphisms

$$[C_{\nabla^1}], [C_{\nabla^0}] : \mathbb{K}[\mathfrak{g}]^G \rightarrow H^*(M, \mathbb{K})$$

agree on all the $\Sigma_k \in \mathbb{K}[\mathfrak{g}]^G$, and by theorem 4.2.4, they agree altogether. $\square$

Finally, we can use the Chern-Weil homomorphism to define the Chern classes of smooth, complex vector bundles.

Definition 4.2.7. Suppose ξ is a smooth, complex vector bundle over a smooth manifold M . Let $P \in \mathbb{C}[\mathfrak{gl}(n, \mathbb{C})]^{\mathrm{GL}(n, \mathbb{C})}$ be the invariant polynomial defined by

$$P(\Omega) = \det\left(I - \frac{i}{2\pi}\Omega\right)$$

The **total Chern class** of ξ is

$$c(\xi) = c_\xi(P) \in H^*(M, \mathbb{C})$$

and the **k -th Chern class** of ξ is the $H^{2k}(M, \mathbb{C})$ -homogeneous component $c_k(\xi)$ of $c(\xi)$.

4.3. Equivalence of two definitions of Chern classes.

Let $f : L \rightarrow M$ be a smooth map, so that we have a smooth pullback bundle $f^*\xi$ over L . Given a connection ∇ on ξ , there is a canonical connection $f^*\nabla$ on $f^*\xi$ given as follows. Let $e = (e_1, \dots, e_n)$ be a local frame on some $U \subseteq M$, and $f^*e = (f^*e_1, \dots, f^*e_n)$ the pullback frame on $f^{-1}(U)$. If $\omega \in \Omega^1(U, \mathfrak{gl}(n, \mathbb{K}))$ denotes the connection form of ∇ relative to e , then $f^*\omega \in \Omega^1(f^{-1}(U), \mathfrak{gl}(n, \mathbb{K}))$, relative to f^*e , defines a connection on $f^{-1}(U)$. One can check that this pullback transforms under change of frame in such a way that makes it independent of e , and hence we get a globally defined connection $f^*\nabla$.

Moreover, one can check that the curvature form of this pullback connection, relative to f^*e , is given by $f^*\Omega$. For any polynomial $P \in \mathbb{C}[\mathfrak{gl}(n, \mathbb{C})]^{\mathrm{GL}(n, \mathbb{C})}$, since $f^* : H^*(M, \mathbb{C}) \rightarrow H^*(L, \mathbb{C})$ is a ring homomorphism, we have $P(f^*\Omega) = f^*P(\Omega) \in H^*(L, \mathbb{C})$. We conclude that the Chern-Weil homomorphism on f^* is given by

$$c_{f^*\xi}(P) = f^*(c_\xi(P))$$

In particular,

Proposition 4.3.1. *The Chern classes satisfy the naturality property*

$$c(f^*\xi) = f^*c(\xi)$$

for any smooth map $f : L \rightarrow M$ and smooth, complex vector bundle ξ over M .

This shows all invariant polynomials $P \in \mathbb{C}[\mathfrak{g}]^G = \mathbb{C}[f_1, \dots, f_n]$ define smooth characteristic classes, and the Chern classes correspond (up to complex factors) to the homogeneous polynomials f_i of degree i .

Secondly, when given vector bundles ξ, ξ' with connections ∇, ∇' , and connection forms ω, ω' relative to frames $e = (e_1, \dots, e_n)$ and $(e'_1, \dots, e'_{n'})$ on $U \subseteq M$, there is a connection $\nabla + \nabla'$ on $\xi \oplus \xi'$ with connection form relative to $(e_1, \dots, e_n, e'_1, \dots, e'_{n'})$ given by the block matrix

$$\omega_{\nabla + \nabla'} = \begin{pmatrix} \omega & 0 \\ 0 & \omega' \end{pmatrix}$$

In the same way, the curvature form of $\nabla + \nabla'$ on $\xi \oplus \xi'$ is given by the block matrix

$$\Omega_{\nabla + \nabla'} = \begin{pmatrix} \Omega & 0 \\ 0 & \Omega' \end{pmatrix}$$

This allows us to prove the following:

Proposition 4.3.2. *The Chern classes satisfy*

$$c(\xi \oplus \xi') = c(\xi) \wedge c(\xi')$$

for any smooth, complex vector bundles ξ, ξ' over M .

Proof. Note that

$$P(\Omega_{\nabla+\nabla'}) = P(\Omega)P(\Omega')$$

hence

$$c(\xi \oplus \xi') = [P(\Omega_{\nabla+\nabla'})] = [P(\Omega)P(\Omega')] = c(\xi) \wedge c(\xi')$$

□

Now by theorem 3.5.11, the topological definition of Chern classes, here denoted by c , are equal to the geometric definition of Chern classes, here denote by $\bar{c}$, up to a scalar factor. We will show they are actually the same.

Now we think of both $c, \bar{c}$ as living in the cohomology groups $H^*(M; \mathbb{C})$ by using the natural homomorphism $H^*(M; \mathbb{Z}) \rightarrow H^*(M; \mathbb{C})$ induced by inclusion $\mathbb{Z} \rightarrow \mathbb{C}$ and the de Rham isomorphism.

For now suppose that ξ is a Riemannian vector bundle of rank 2 over a closed surface M , and choose a connection ∇ on ξ compatible with the metric. Then, with respect to a local, positively oriented orthonormal frame the connection form is skew-symmetric:

$$\omega = \begin{pmatrix} 0 & \omega_1^2 \\ -\omega_1^2 & 0 \end{pmatrix}$$

and so is $\Omega = d\omega$, since $\omega_1^2 \wedge \omega_1^2 = 0$:

$$\Omega = \begin{pmatrix} 0 & \Omega_1^2 \\ -\Omega_1^2 & 0 \end{pmatrix}$$

where $\Omega_1^2 = d\omega_1^2 \in H^2(M, \mathbb{R}) \subseteq H^2(M, \mathbb{C})$. Note that Ω_1^2 is independent of the choice of local frame, and therefore defined globally. However, ω_1^2 still depend on the local frames and hence Ω_1^2 is not necessarily (globally) exact.

Theorem 4.3.3 (Gauss-Bonnet). *On any oriented Euclidean vector bundle ξ of rank 2 over a manifold M , the Euler class can be represented as*

$$e(\xi) = \frac{1}{2\pi}[\Omega_1^2]$$

In classical Riemannian geometry, one considers the tangent bundle τ_M of a Riemannian manifold M of dimension 2. Writing $\Omega_1^2 = \kappa dA$, κ is known as the **Gaussian curvature**, and by theorem 3.3.13 and 4.3.3, we have

Corollary 4.3.4 (Classical Gauss-Bonnet). *On any closed, orientable surface M , equipped with a Riemannian metric, with Gaussian curvature κ , we have*

$$\int_M \kappa dA = 2\pi\chi(M)$$

Proof. (of theorem 4.3.3) Orientability together with the Euclidean metric on ξ allows us to define a complex structure on ξ , turning it into a complex line bundle $\xi_{\mathbb{C}}$. Choosing a compatible connection ∇ on (the Euclidean bundle) ξ , we have

$$\begin{aligned} \nabla s_1 &= \nabla_1^2 \otimes s_2 \\ \nabla s_2 &= -\omega_1^2 \otimes s_1 \end{aligned}$$

This gives us a connection $\nabla_{\mathbb{C}}$ on the complex line bundle $\xi_{\mathbb{C}}$, reading

$$\nabla_{\mathbb{C}} s = i\omega_1^2 \otimes s$$

But now we are in complex rank 1, so our connection and curvature matrices consist only of one entry: the connection form is $i\omega_1^2$, and the complex curvature form is therefore $i\Omega_1^2$. The first Chern class of $\xi_{\mathbb{C}}$ is thus given by the degree-2-part of

$$\bar{c}(\xi_{\mathbb{C}}) = [1 - \frac{i}{2\pi}(i\Omega_1^2)] = [1 + \frac{1}{2\pi}\Omega_1^2]$$

which is

$$\bar{c}_1(\xi_{\mathbb{C}}) = [\frac{1}{2\pi}\Omega_1^2]$$

Both $\bar{c}_1$, as well as c_1 , which is equal to the Euler class, are characteristic classes for complex line bundles, so they have to agree up to a complex scalar factor:

$$\bar{c}_1(\xi_{\mathbb{C}}) = \alpha e(\xi), \alpha \in \mathbb{C}$$

Now we pick the Euclidean tangent bundle τ on S^2 , and view it as a complex vector bundle in the way above. The unit sphere S^2 has Gaussian curvature 1, so Ω_1^2 integrates to the area 4π . On the other hand, by theorem 3.3.13, $e(\tau)$ integrates to the Euler characteristic:

$$\int_{S^2} \bar{c}_1(\tau_{\mathbb{C}}) = 2 = \chi(S^2) = \int_{S^2} e(\tau)$$

so $\alpha = 1$. □

In particular, the above argument shows that the $\bar{c}_1$ always coincides with c_1 . Thus we conclude that:

Theorem 4.3.5. *On any smooth, complex vector bundle ξ , the geometric definition of the Chern class coincides with the topological definition:*

$$\bar{c}(\xi) = c(\xi) \in H^*(M, \mathbb{C})$$

We give a final remark: in fact, Chern-Weil theory can be applied to any **principal G -bundle**, not only vector bundles. (Vector bundles are special principal G -bundles.) However, we may not have the corresponding topological definition of characteristic classes for principal G -bundles. See [26] for more detail.

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